



Melitech CRM

Comprehensive Platform Documentation

Enterprise Customer Relationship Management, Accounting, HR,
Procurement, Multi-Tenancy & Business Intelligence Platform

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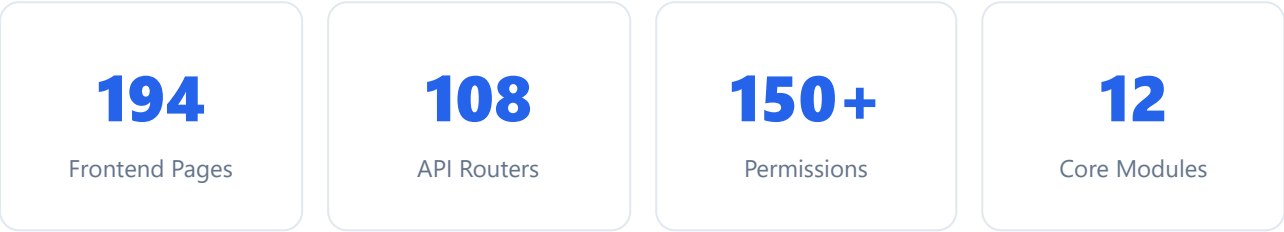
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1. Platform Overview

Melitech CRM is an enterprise-grade, multi-tenant Customer Relationship Management platform built for organisations operating across East Africa and beyond. It unifies client management, invoicing, accounting, human resources, procurement, project management, communications, and business intelligence into a single, cohesive web application.



Key Capabilities

Client & Contact Management

Full lifecycle client records, custom fields, organisation linkages, activity timelines, and bulk operations.

Sales & Revenue

Invoices, recurring billing, estimates, quotations, sales pipeline, opportunities, credit/debit notes, and receipts.

Accounting & Finance

Chart of Accounts, payments, expenses, budgets, bank reconciliation, tax compliance, and financial forecasting.

Human Resources

Employee management, departments, attendance, payroll with Kenyan tax engine, leave management, job groups, performance reviews.

Procurement

Suppliers, LPOs, purchase orders, GRNs, inventory management, delivery notes, and imprests.

Project Management

Projects with milestones, work orders, time tracking, service invoices, and project analytics.

Multi-Tenancy

Enterprise multi-organisation support with tenant isolation, pricing tiers, feature flags, and centralised communications.

Business Intelligence

Advanced reports, KPI tracking, report builder, revenue forecasting, conversion analytics, and real-time dashboards.

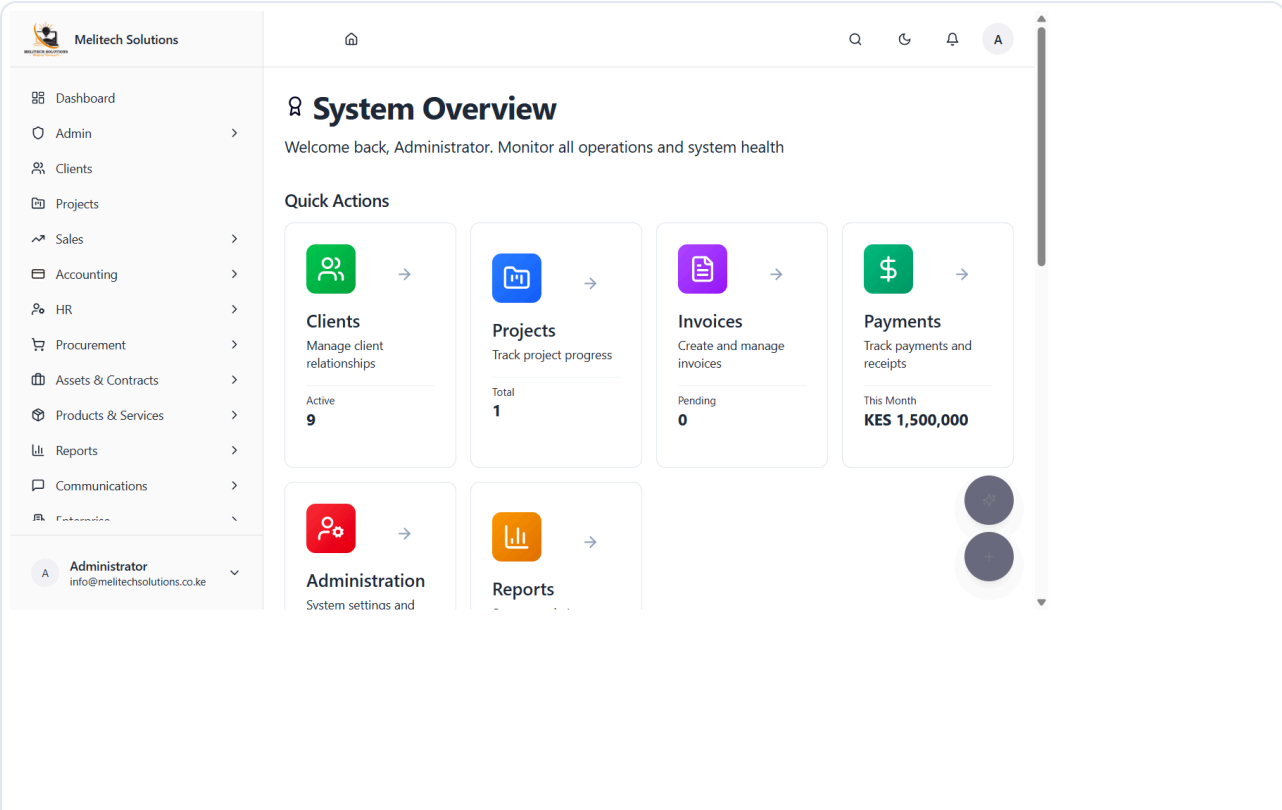


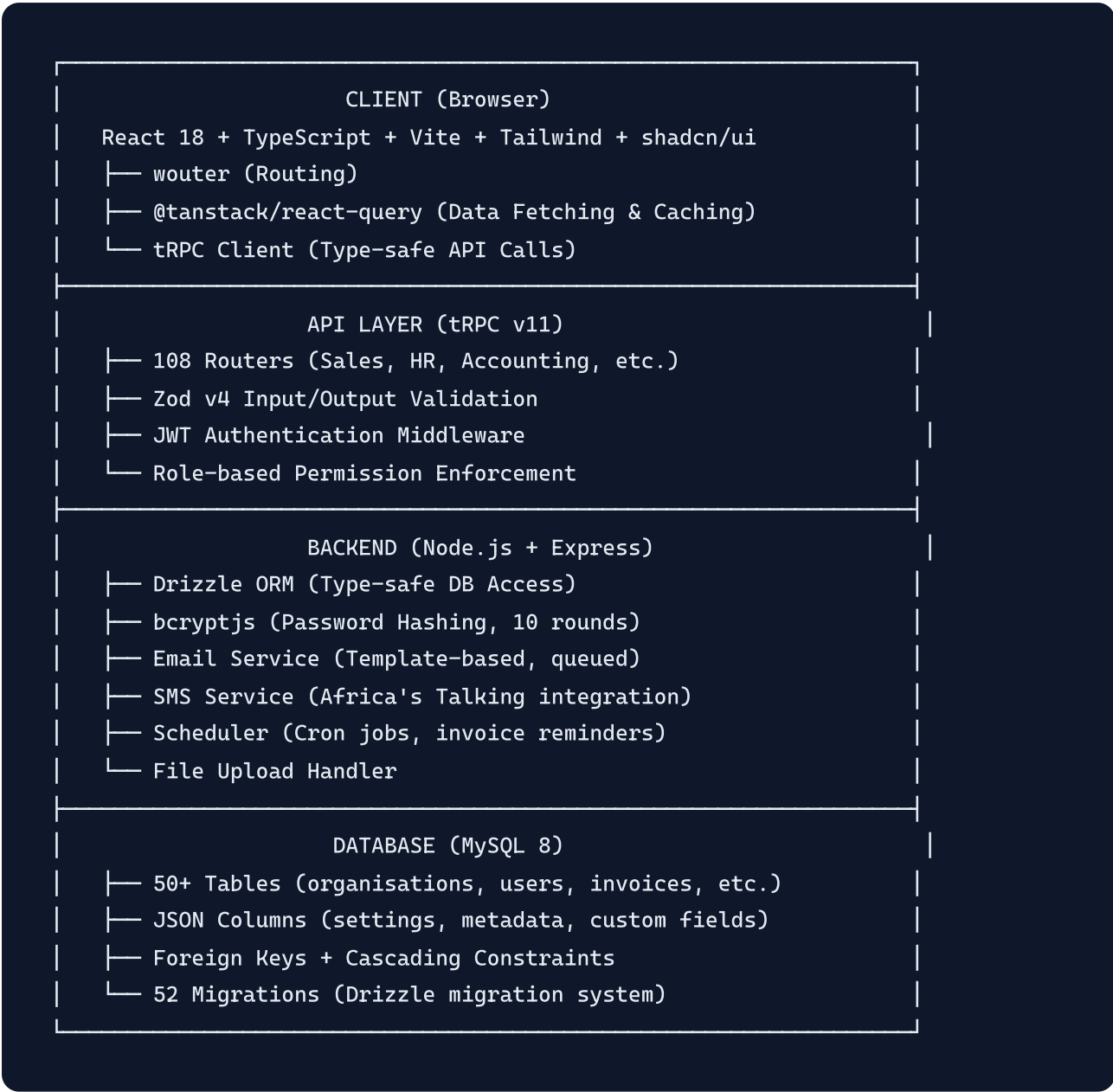
Figure 1.1 — Melitech CRM Landing Page

2. Technical Architecture

Technology Stack

Layer	Technology	Details
Frontend	React 18 + TypeScript	Single Page Application with Vite build tool
UI Framework	Tailwind CSS + shadcn/ui	Radix primitives, custom theme engine
Charts	Recharts	Interactive graphs, dashboards, analytics
API Layer	tRPC v11	End-to-end type-safe RPC with Zod validation
Backend	Node.js + Express	REST + tRPC hybrid, JWT authentication
ORM	Drizzle ORM	Type-safe MySQL queries, migration system
Database	MySQL 8	utf8mb4 collation, JSON columns, full-text search
Payments	Stripe + M-Pesa	Card payments, STK push mobile money
Icons	Lucide React	Consistent icon set across all modules
Deployment	Docker	Multi-container with docker-compose

System Architecture



Database Entity Relationship Overview

Entity Group	Tables	Key Relationships
Authentication	users, sessions, password_resets	users → organizations
CRM	clients, contacts, activities, custom_fields	clients → organizations, contacts → clients
Sales	invoices, invoice_items, estimates, opportunities, receipts, credit_notes, debit_notes	invoices → clients, items → invoices
Accounting	payments, expenses, chart_of_accounts, budgets, bank_reconciliation	payments → invoices, expenses → categories
HR	employees, departments, attendance, payroll, leave_requests, job_groups	employees → departments, payroll → employees
Procurement	suppliers, lpos, grn, inventory, delivery_notes	lpos → suppliers, grn → lpos
Projects	projects, milestones, work_orders, time_tracking	milestones → projects, work_orders → projects
Multi-Tenancy	organizations, organization_features, pricing_tier_features, tenant_messages	org_features → organizations, users → organizations

3. Dashboard & Navigation

Melitech CRM provides **role-specific dashboards** tailored to each user type. Upon login, users are automatically routed to their personalised dashboard with relevant metrics, quick actions, and module shortcuts.

Available Dashboards

Role	Dashboard Route	Key Widgets
Super Admin	/crm/super-admin	Organisation stats, system health, user metrics, revenue overview, all modules
Admin	/crm/admin	User management, approvals, full module access
HR Manager	/crm/hr	Employee count, attendance today, leave requests, payroll summary
Accountant	/crm/accountant	Revenue overview, outstanding invoices, expense tracking, P&L
Project Manager	/crm/project-manager	Active projects, milestones, team workload, deadlines
Procurement	/crm/procurement	Pending LPOs, inventory alerts, supplier metrics
Sales Manager	/crm/sales	Pipeline value, conversion rates, top clients, forecasts
ICT Manager	/crm/ict	System health, tickets, integrations status
Staff	/crm/staff	My tasks, attendance, leave balance, team chat

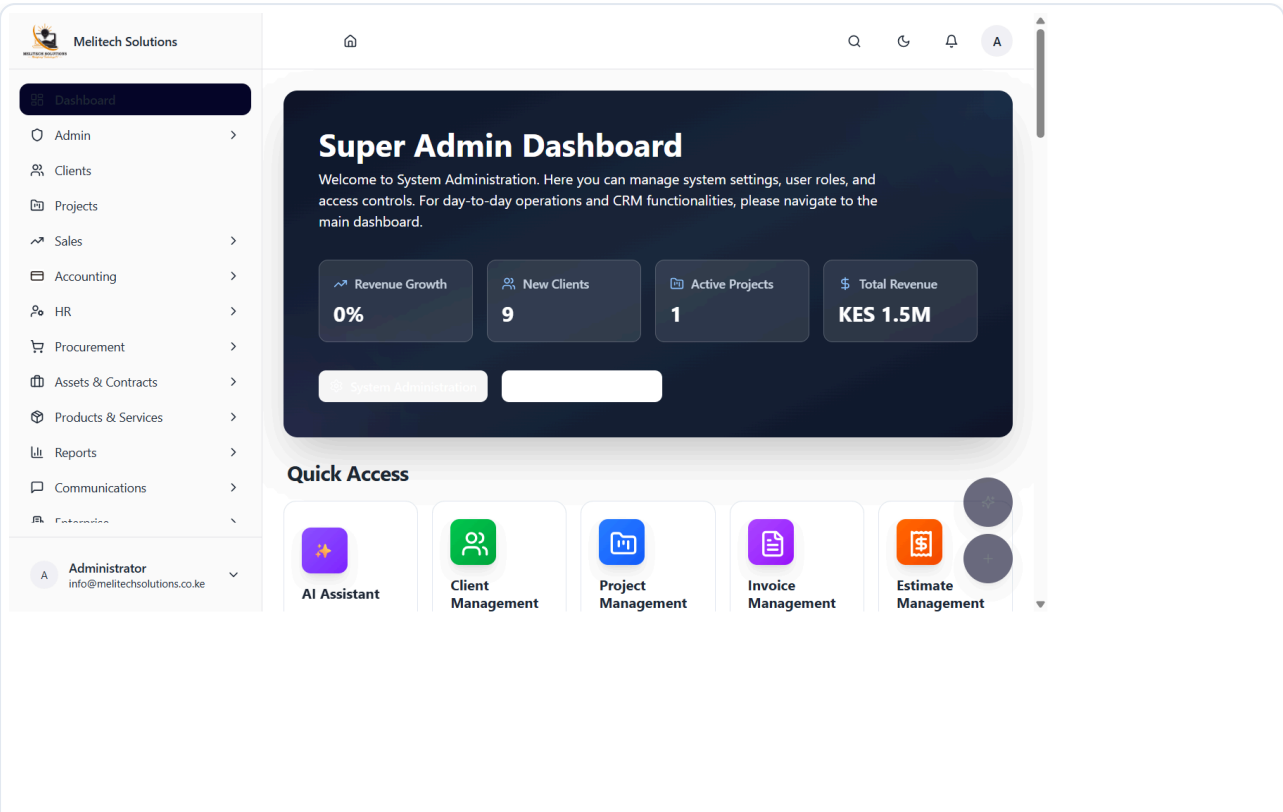


Figure 3.1 — Super Admin Dashboard with system-wide metrics and quick actions

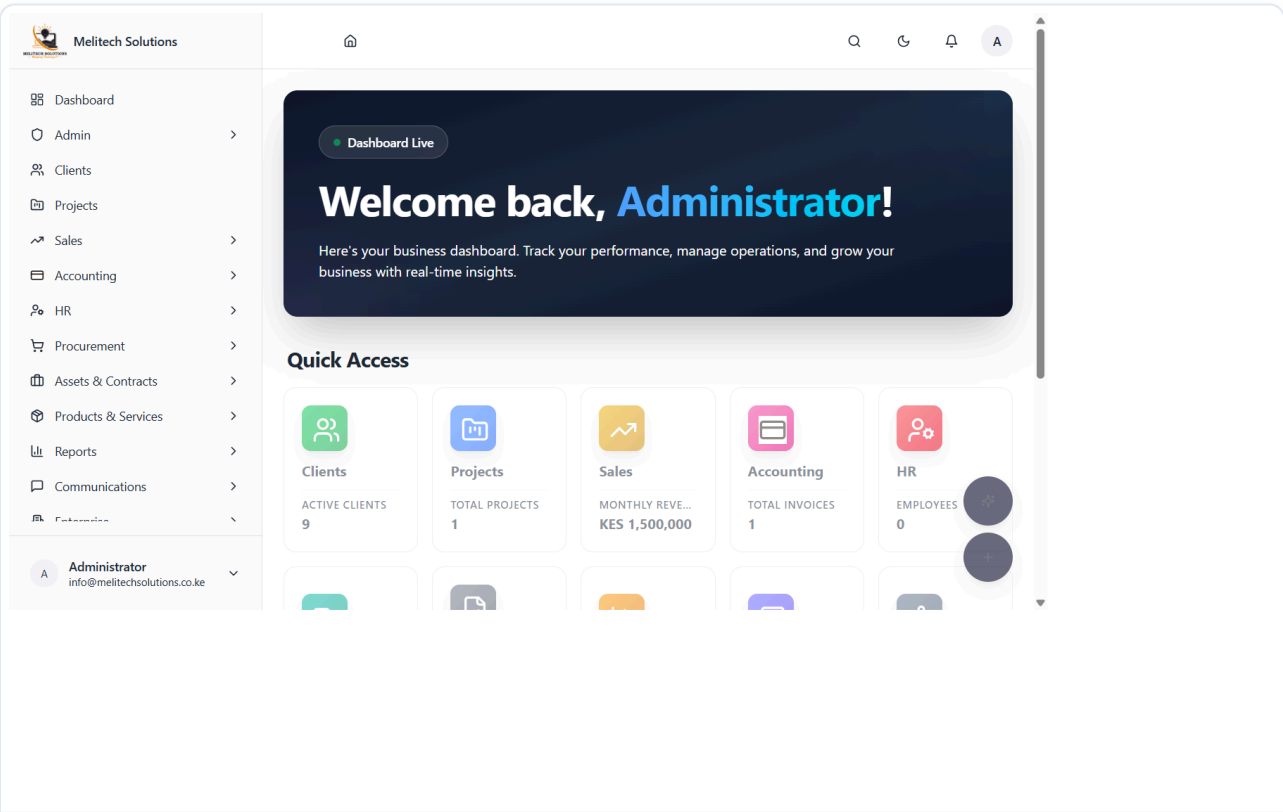


Figure 3.2 — Main CRM Dashboard with module shortcuts, KPI widgets, and recent activity feed

Sidebar Navigation

The collapsible sidebar adapts to the user's role, showing only modules the user has permission to access. Super admins see the full navigation with 14 top-level sections and 80+ sub-items.

Navigation Sections (Super Admin View)

Section	Sub-Items	Description
Dashboard	1	Role-specific home dashboard
Admin	2	User management, approvals
Clients	1	Client directory and management
Projects	1	Project management hub
Sales	11	Invoices, estimates, quotations, receipts, pipeline, credit/debit notes
Accounting	11	Payments, expenses, CoA, reconciliation, budgets, tax, billing
HR	7	Employees, departments, attendance, payroll, leave, reviews
Procurement	7	Suppliers, LPOs, orders, imprests, inventory, delivery, GRNs
Assets & Contracts	4	Assets, contracts, warranty, work orders
Products & Services	3	Products, services, service templates
Reports	9	Financial, sales, HR, KPI, report builder, activity trail
Communications	5	Email, tickets, notifications, team chat, AI assistant
Enterprise	5	Multi-tenancy, tenant admins, pricing, comms, settings
Tools	14	System settings, backups, themes, custom fields, workflows, integrations

4. CRM — Client Management

The CRM module is the core of client relationship tracking. It provides a comprehensive client registry with full contact details, activity history, associated invoices, projects, and customisable fields.

Features

- **Client Directory** — Searchable, paginated list with bulk operations and CSV/PDF export
- **Client Profiles** — Detailed view with tabs for contacts, invoices, projects, payments, activities
- **Custom Fields** — 14+ field types (text, number, date, dropdown, checkbox, etc.) per entity
- **Advanced Filtering** — Dynamic filters with saved filter presets
- **Bulk Operations** — Multi-select with batch status updates, export, and delete
- **Activity Timeline** — Complete audit trail of all client interactions
- **Organisation Linking** — Associate clients with parent companies or groups
- **Data Import** — Excel/CSV bulk import with field mapping

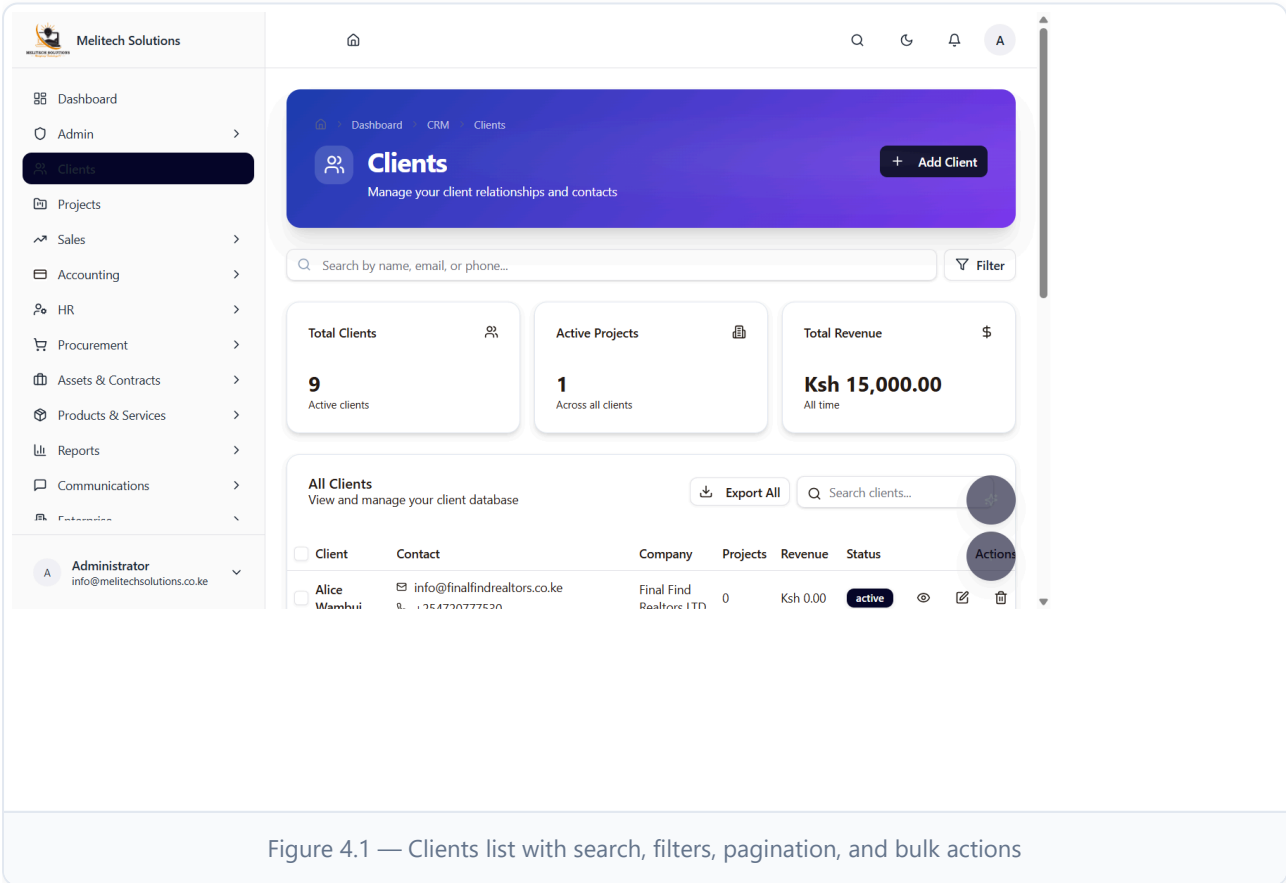


Figure 4.1 — Clients list with search, filters, pagination, and bulk actions

Client Data Model

Field	Type	Description
Name	Text	Client or company name
Email	Email	Primary contact email
Phone	Phone	Contact phone number
Company	Text	Associated company name
Status	Enum	Active, Inactive, Lead, Prospect
Country / Address	Text	Physical location details
KRA PIN	Text	Tax identification (Kenya)
Custom Fields	Dynamic	User-defined fields with validation

5. Sales & Invoicing

The Sales module covers the complete revenue lifecycle from opportunity identification through quotation, invoicing, payment collection, and receipt generation.

5.1 Invoices

Create, send, and track invoices with automatic tax calculation, line item management, payment tracking and overdue alerts.

- Professional invoice templates with company branding
- Automatic invoice numbering with customisable prefixes
- Multi-currency support
- Partial payment tracking
- Overdue invoice alerts and reminders (automated via scheduler)
- PDF generation and email delivery
- Recurring invoices with configurable intervals

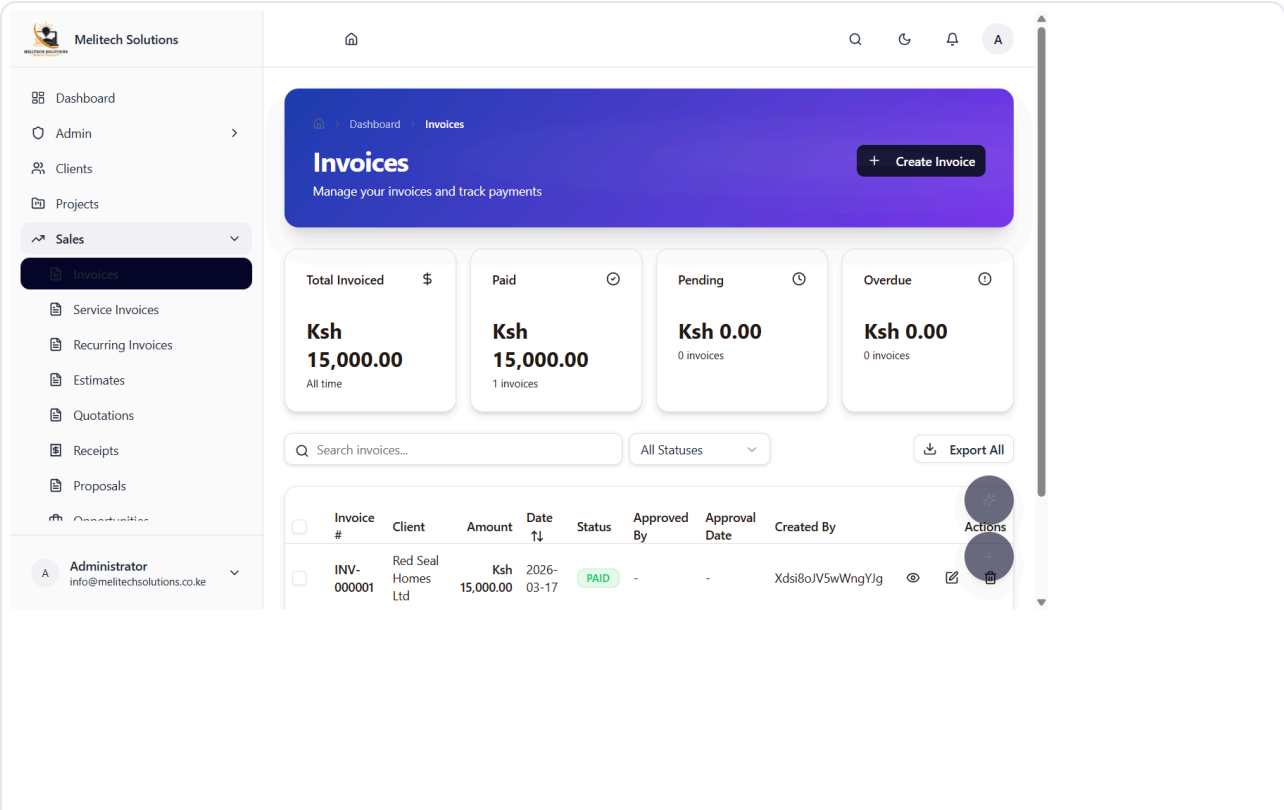


Figure 5.1 — Invoice management with status tracking, search, and export

5.2 Sales Pipeline & Opportunities

Visual Kanban-style pipeline for tracking deals through stages. Each opportunity records value, probability, expected close date, and associated contacts.

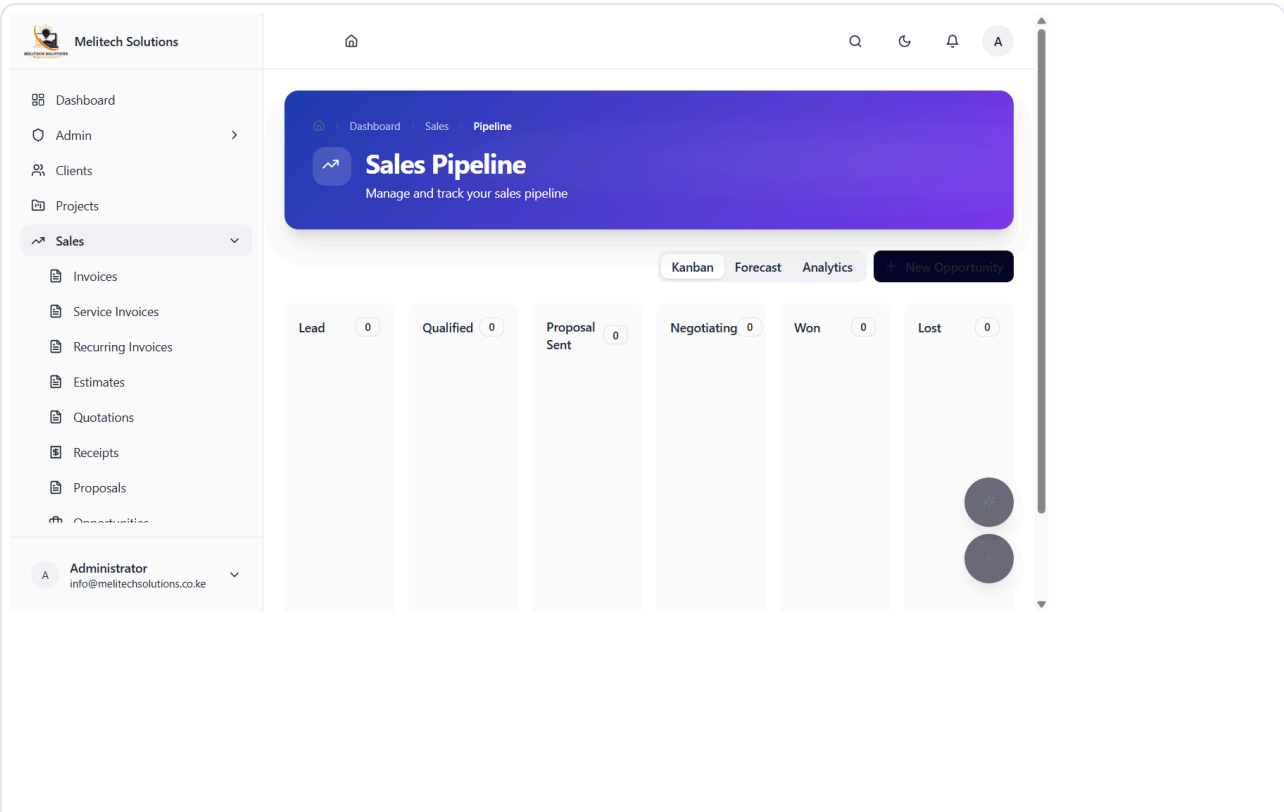


Figure 5.2 — Sales pipeline with Kanban-style deal management and stage progression

5.3 Estimates & Quotations

Generate professional estimates that can be converted to invoices with a single click. Track estimate status (Draft → Sent → Accepted → Converted).

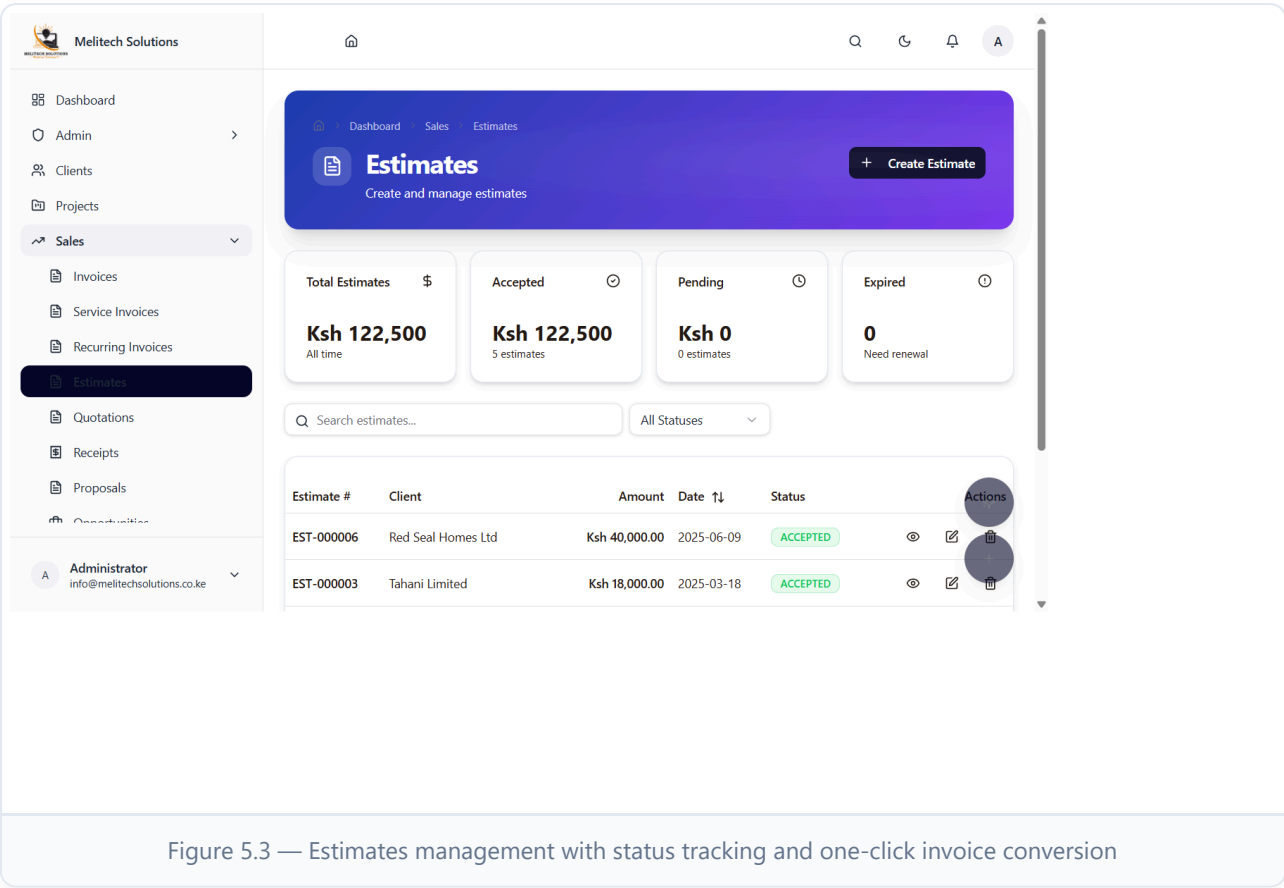


Figure 5.3 — Estimates management with status tracking and one-click invoice conversion

5.4 Credit & Debit Notes

Issue credit notes for returns or adjustments, and debit notes for additional charges. Both link back to the original invoice for audit trail.

5.5 Receipts

Auto-generated receipts upon payment confirmation. Supports manual receipt creation with configurable templates.

Sales Module Summary

Sub-Module	Route	Key Actions
Invoices	/invoices	Create, Edit, Send, Track Payments, Export
Service Invoices	/service-invoices	Service-based billing with templates
Recurring Invoices	/recurring-invoices	Automated periodic billing
Estimates	/estimates	Create, Send, Convert to Invoice
Quotations	/quotations	Formal price quotes for clients
Receipts	/receipts	Payment confirmation documents
Proposals	/proposals	Business proposal generation
Opportunities	/opportunities	Deal tracking with pipeline stages
Sales Pipeline	/sales-pipeline	Visual Kanban deal management
Credit Notes	/credit-notes	Client credit adjustments
Debit Notes	/debit-notes	Additional charge documents

6. Accounting & Finance

A full-featured accounting module with double-entry bookkeeping, bank reconciliation, budgeting, and Kenyan tax compliance built in.

6.1 Chart of Accounts

Hierarchical account structure supporting Assets, Liabilities, Equity, Revenue, and Expense categories with sub-accounts.

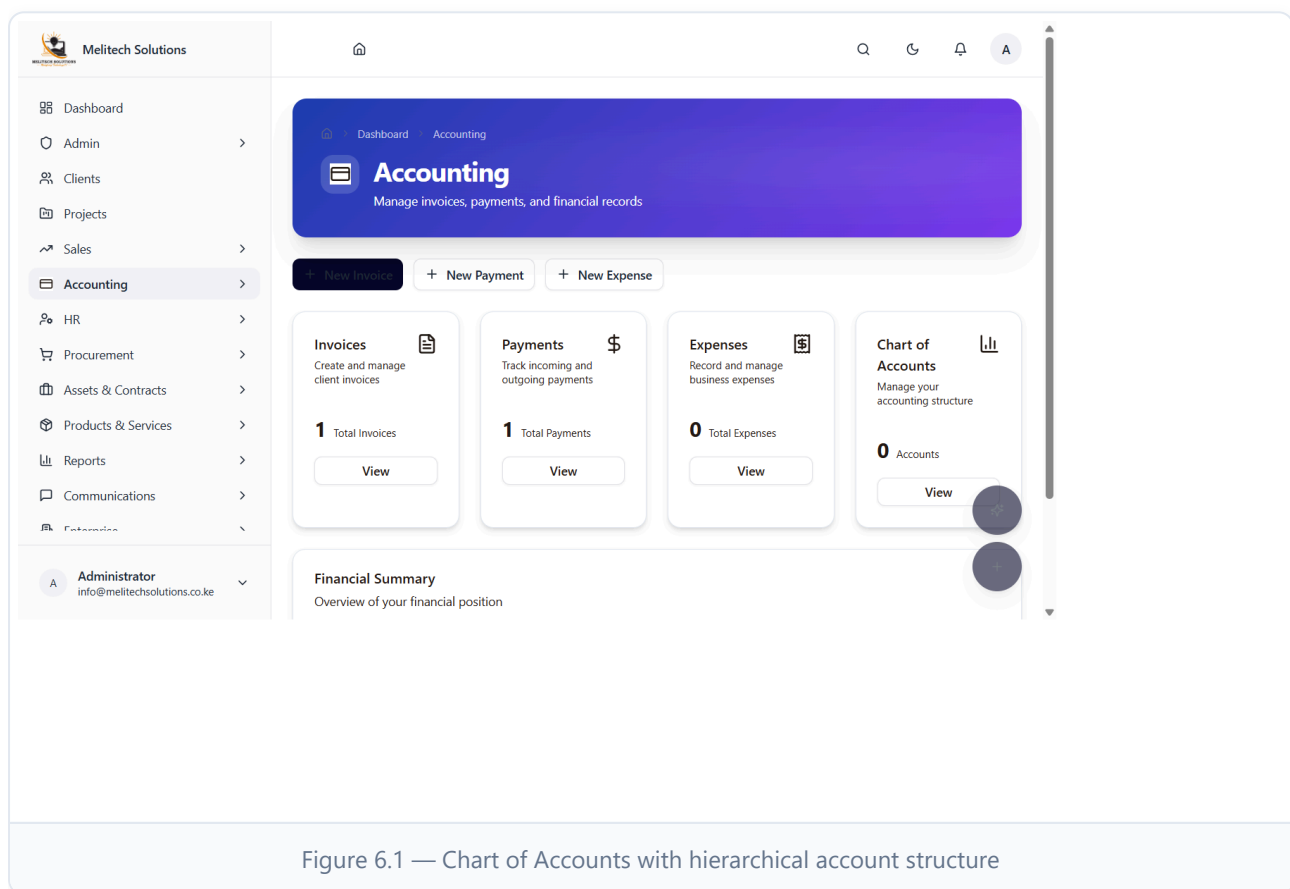


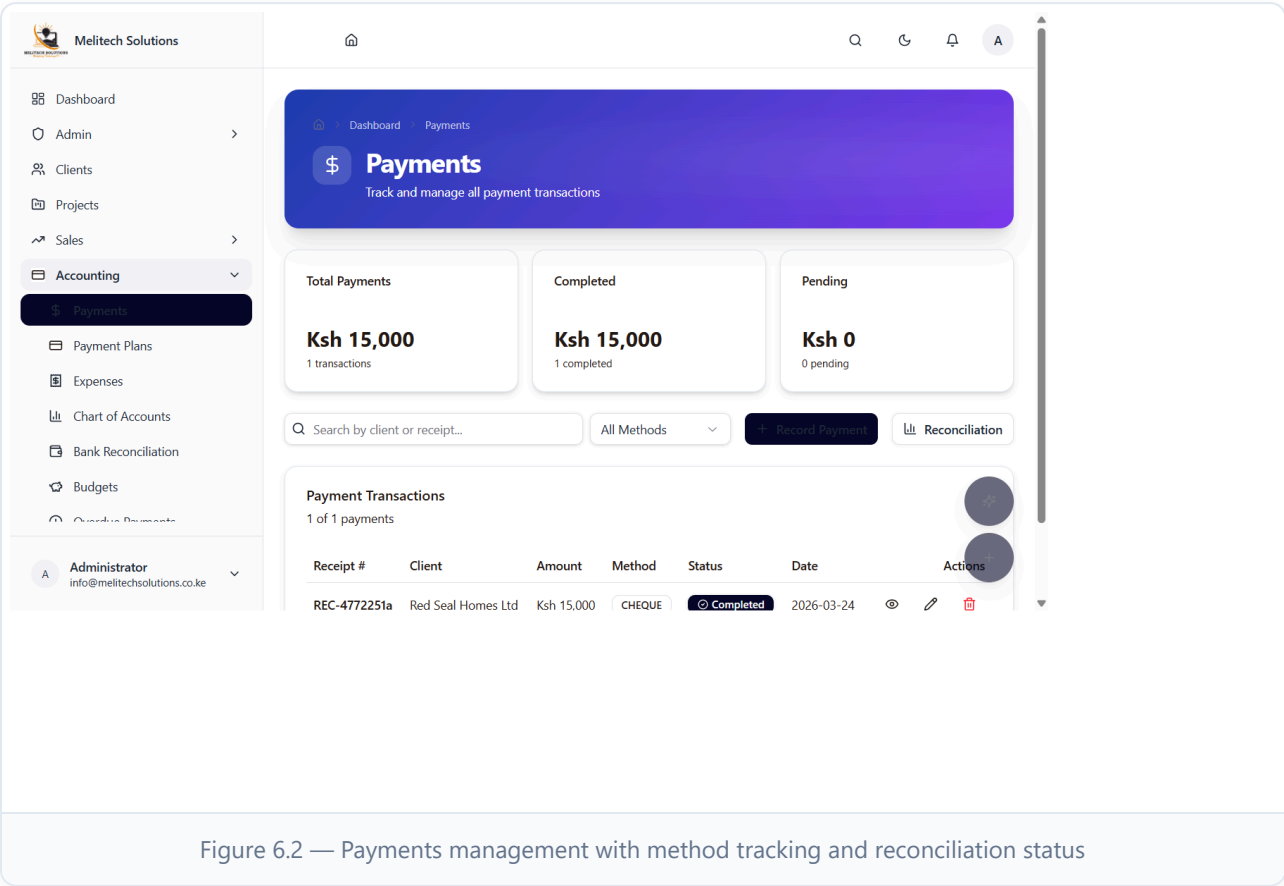
Figure 6.1 — Chart of Accounts with hierarchical account structure

6.2 Payments

Record and track all incoming and outgoing payments with multiple payment method support (Bank Transfer, M-Pesa, Stripe, Cash, Cheque).

- Payment recording with invoice linking
- Partial payment support
- Overdue payment dashboard

- Payment reconciliation with bank statements
- Payment plans for instalment billing



6.3 Expenses

Track business expenses by category, department, and project. Supports receipt uploads, approval workflows, and expense reports.

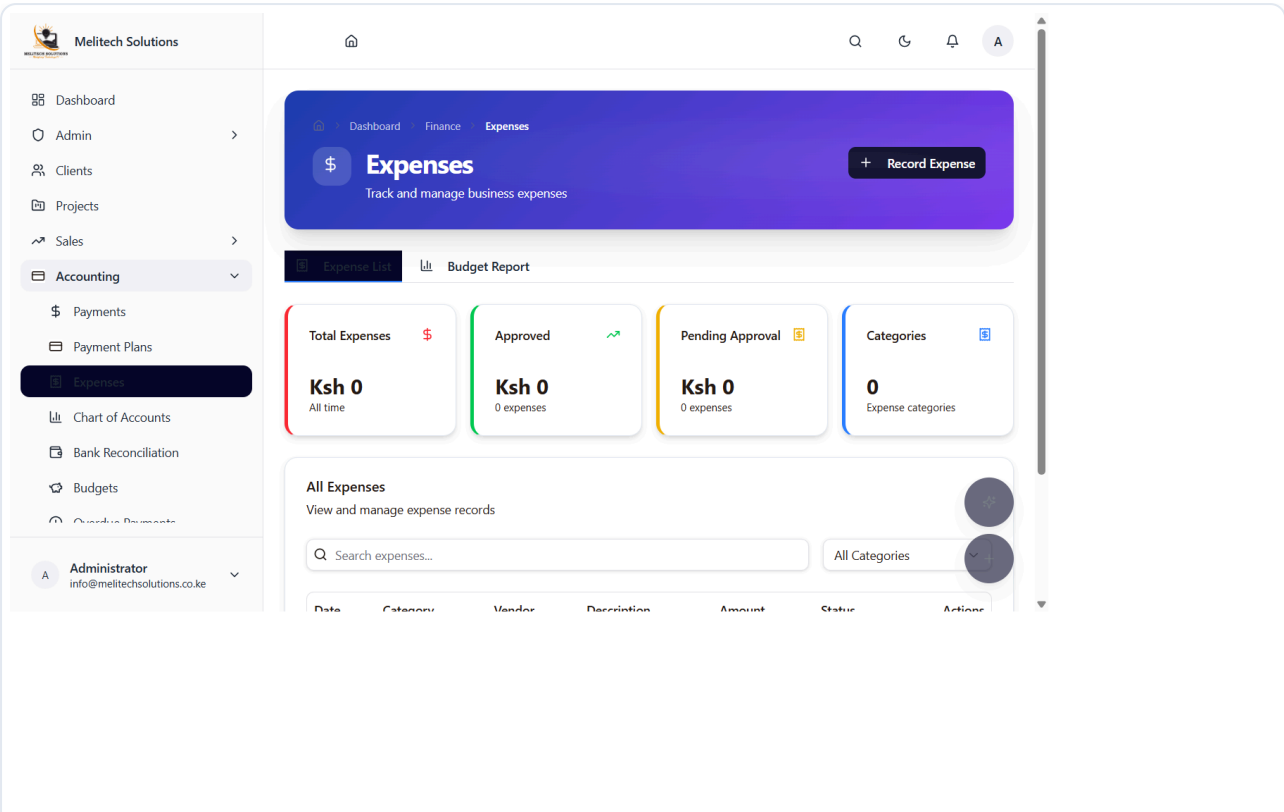


Figure 6.3 — Expense tracking with category-based organisation

6.4 Budgets

Create department and project budgets, track actual spend vs. budget, and receive alerts when thresholds are exceeded.

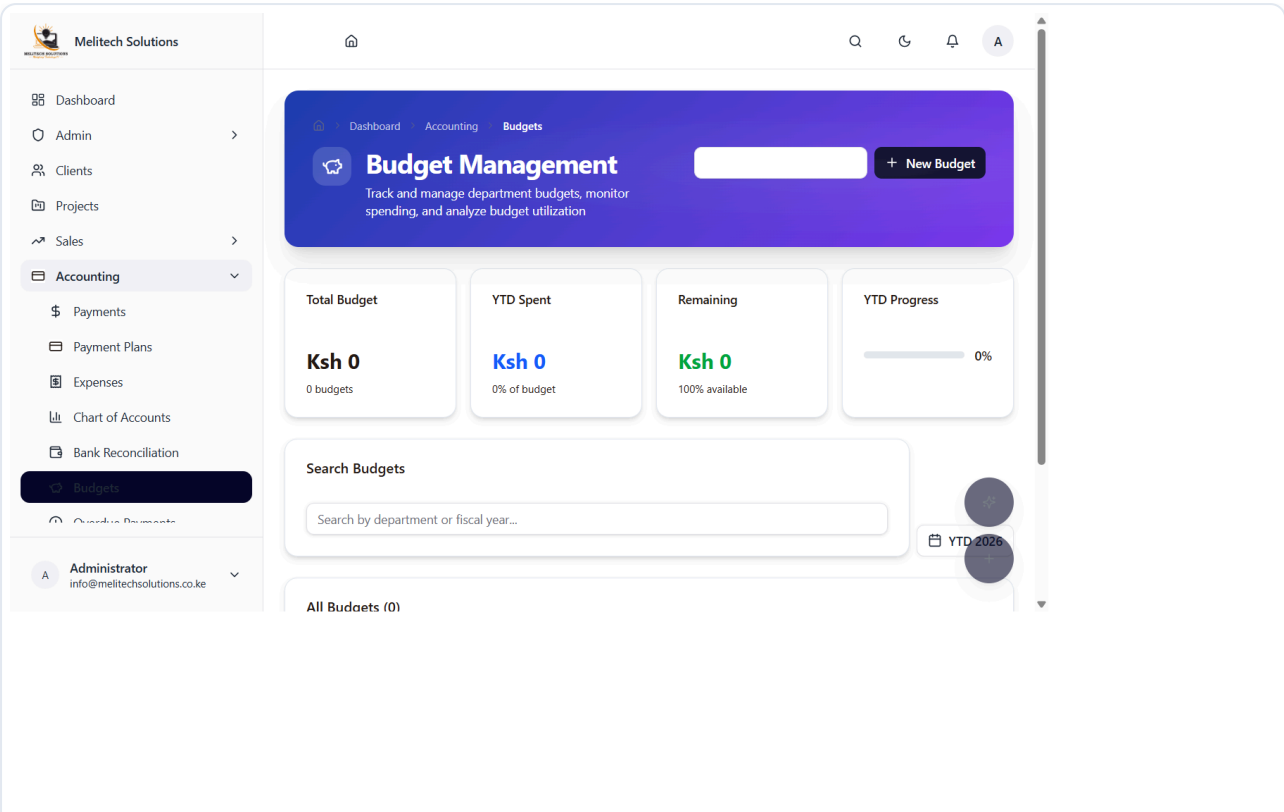


Figure 6.4 — Budget management with variance tracking and departmental breakdown

6.5 Financial Tools

Tool	Route	Description
Bank Reconciliation	/bank-reconciliation	Match bank transactions with system records
Budgets	/budgets	Department/project budget creation and tracking
Tax Compliance	/payroll/tax-compliance	KRA tax reports, P9 forms, PAYE summaries
Financial Dashboard	/financial-dashboard	Real-time P&L, cash flow, revenue charts
Forecasting	/forecasting	Revenue and expense projections
Billing	/billing	Subscription and recurring charge management
Overdue Payments	/payments/overdue	Aged receivables with follow-up actions
Payment Plans	/payment-plans	Instalment schedules and tracking

7. Human Resources

End-to-end HR management from recruitment through payroll processing. Includes Kenyan payroll compliance with PAYE, NHIF, NSSF, and Housing Levy calculations.

7.1 Employees

Comprehensive employee records with personal details, employment history, documents, and role assignments.

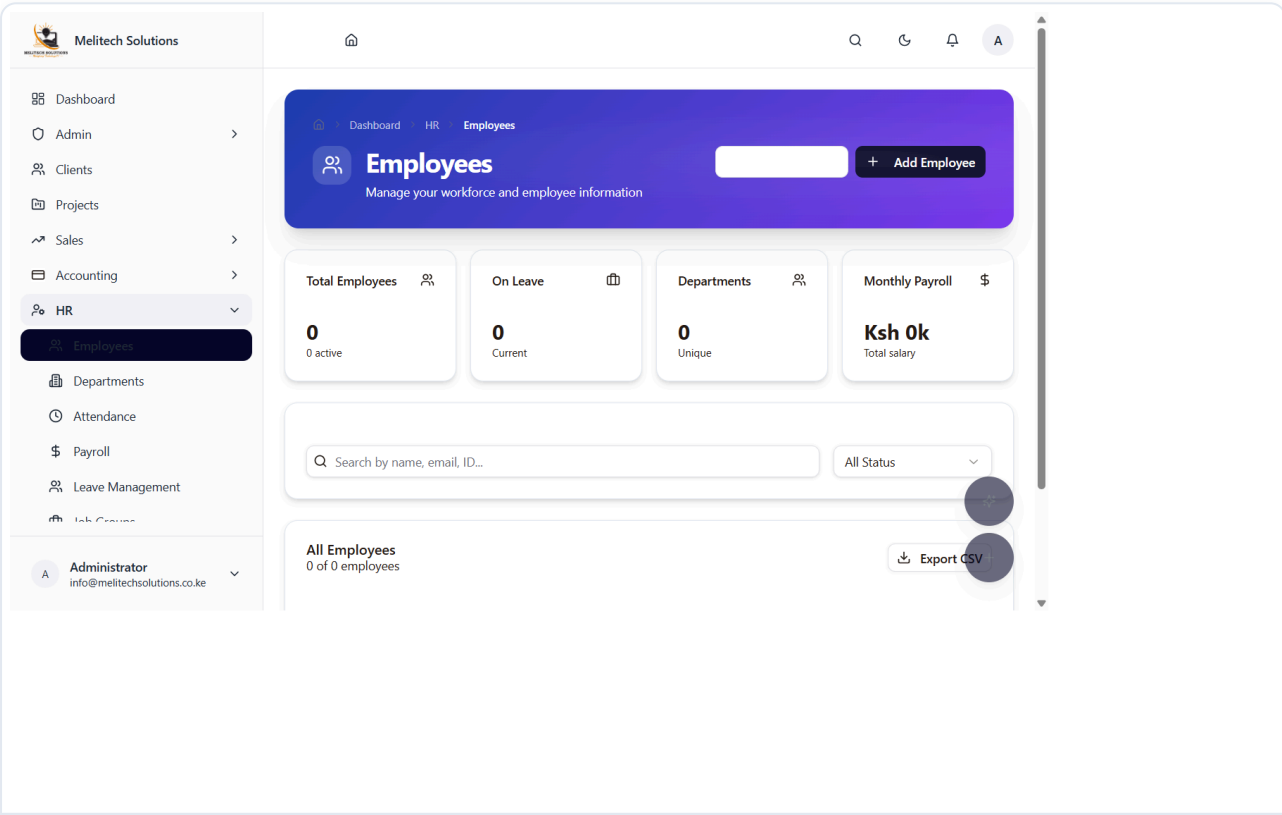


Figure 7.1 — Employee directory with department and status filtering

7.2 Attendance

Daily attendance tracking with clock-in/clock-out, overtime calculation, and attendance analytics.

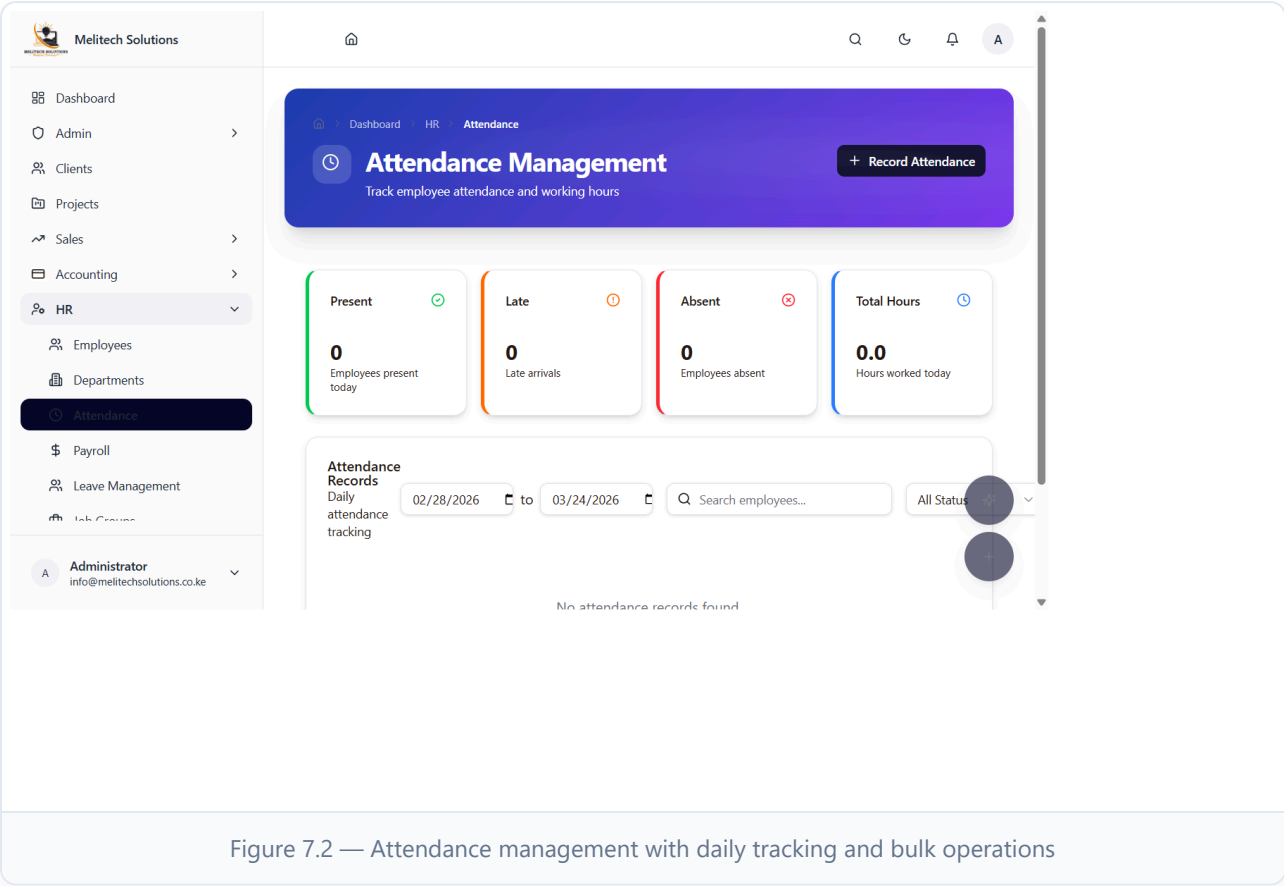


Figure 7.2 — Attendance management with daily tracking and bulk operations

7.3 Payroll

Full payroll processing with automatic statutory deduction calculations for Kenyan employment law.

Kenyan Payroll Compliance: Automatic calculation of PAYE (progressive tax bands), NHIF contributions, NSSF Tier I/II, Housing Levy (1.5%), and employer contributions.

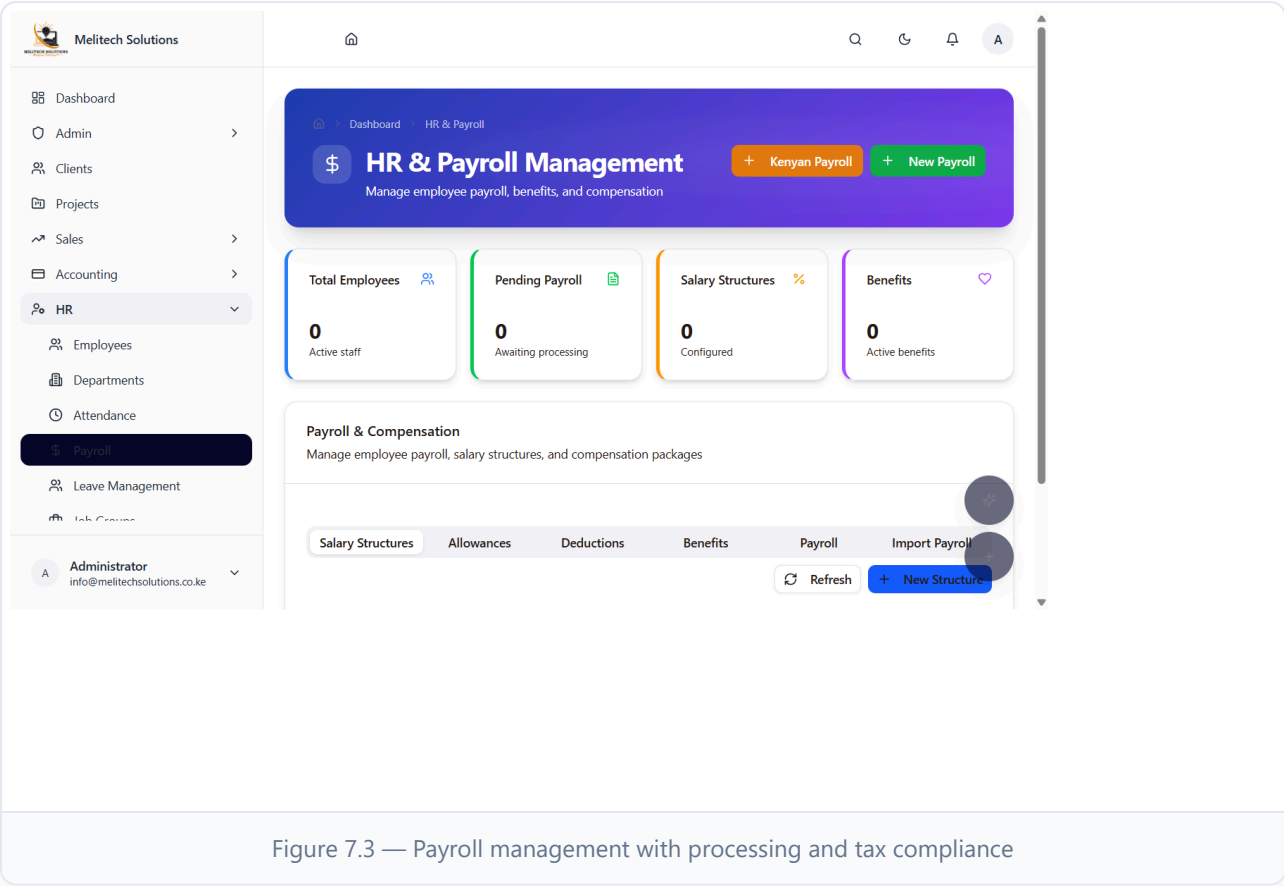


Figure 7.3 — Payroll management with processing and tax compliance

7.4 Leave Management

Configure leave types (annual, sick, maternity, paternity, compassionate), track balances, and process approval workflows.

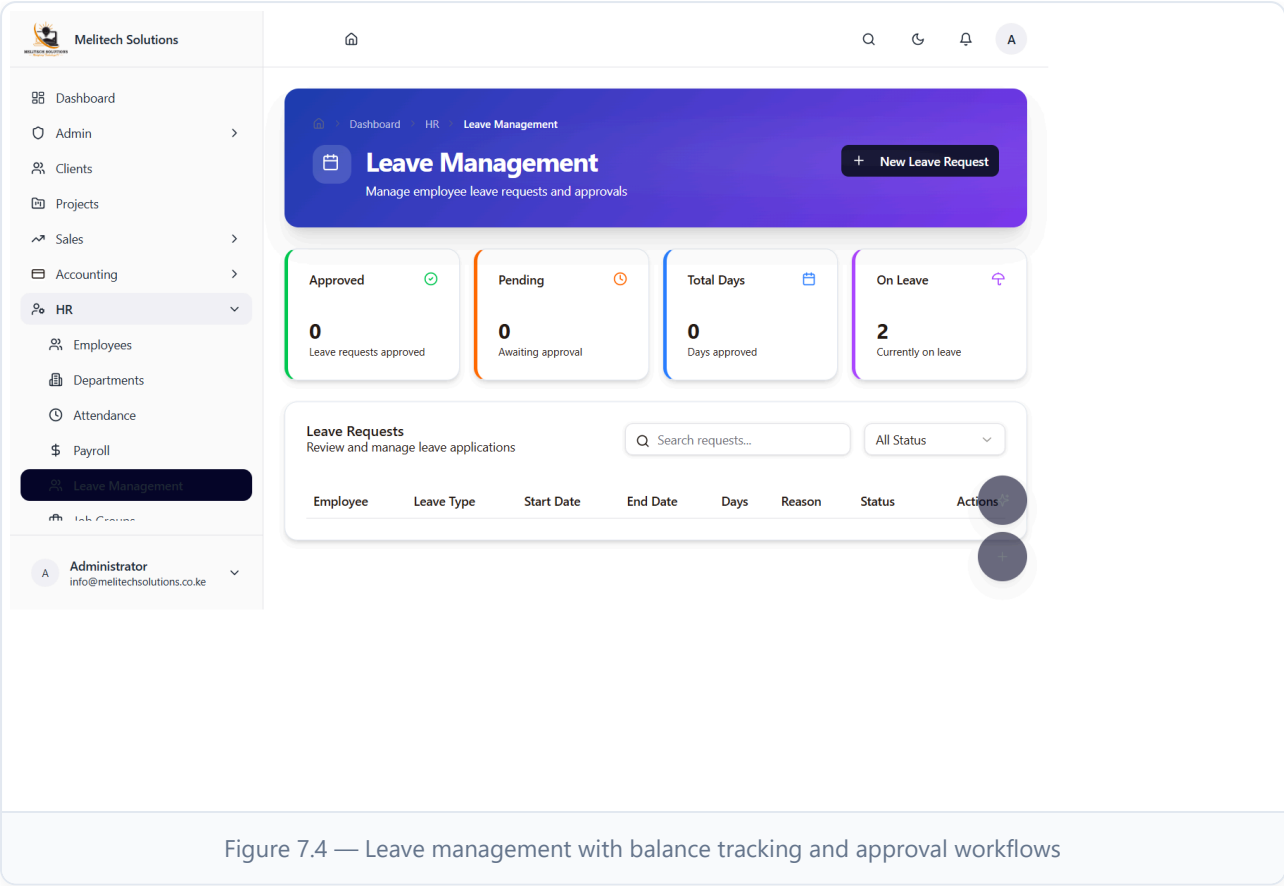


Figure 7.4 — Leave management with balance tracking and approval workflows

7.5 Departments

Manage organisational departments with head assignments, budget allocations, and employee counts per department.

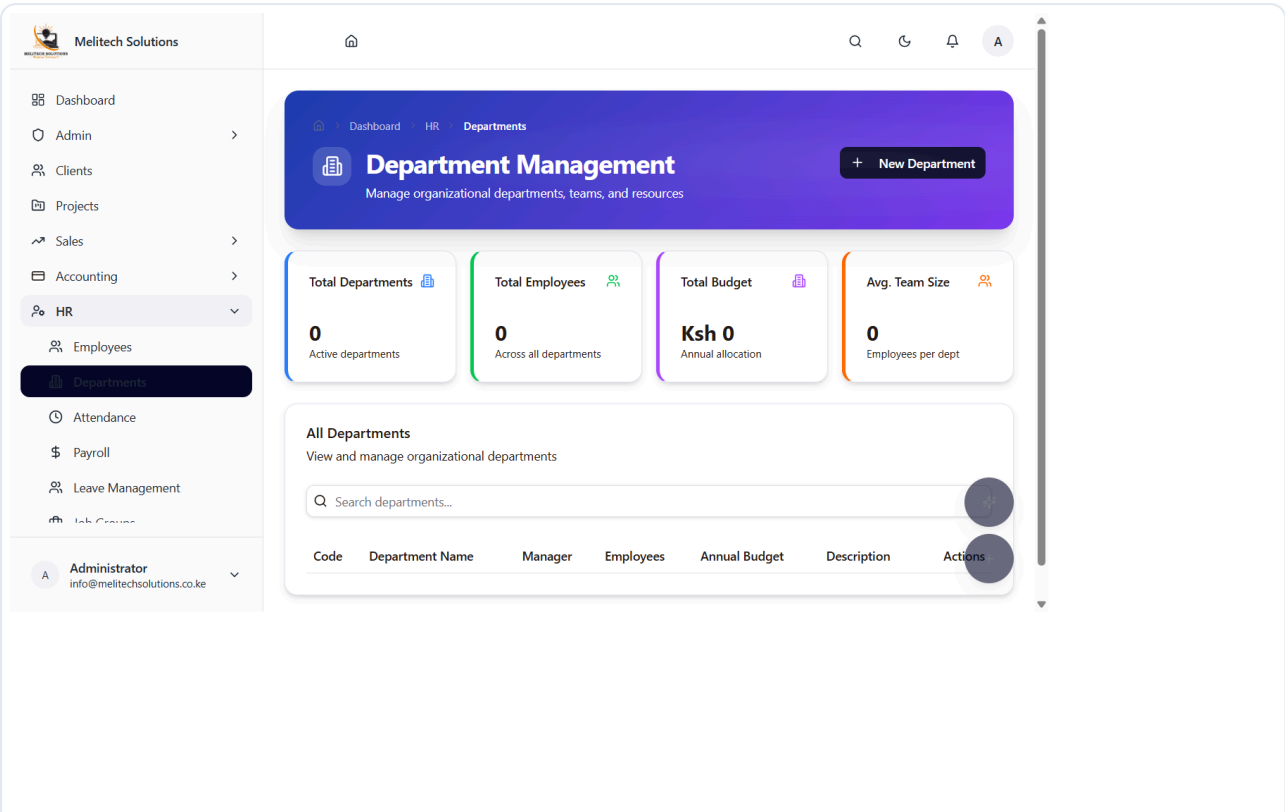


Figure 7.5 — Department management with head assignments and headcount tracking

HR Module Summary

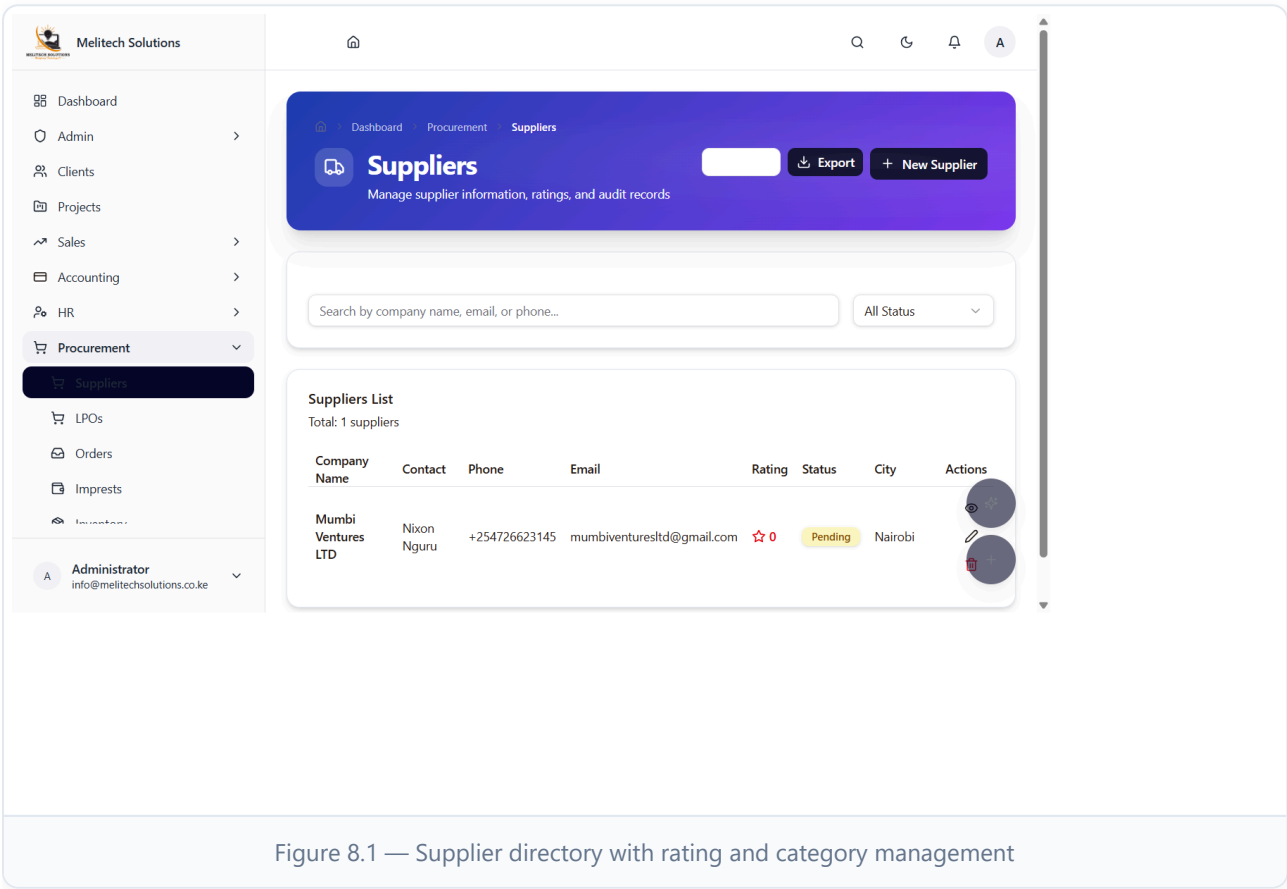
Sub-Module	Route	Key Features
Employees	/employees	CRUD, document uploads, role assignment
Departments	/departments	Department hierarchy, head assignment
Attendance	/attendance	Clock in/out, overtime, reports
Payroll	/payroll	Processing, KE calculator, payslips
Leave Management	/leave-management	Requests, approvals, balances
Job Groups	/job-groups	Grades, salary structures, bands
Performance Reviews	/performance-reviews	Review cycles, goals, ratings

8. Procurement & Inventory

Manage the complete procurement lifecycle from supplier management through purchase ordering, goods receipt, and inventory tracking.

8.1 Suppliers

Maintain a supplier registry with contact details, payment terms, rating scores, and purchase history.



8.2 Procurement Hub

Centralised procurement command centre with overview of pending orders, supplier performance, LPO status, and inventory alerts.

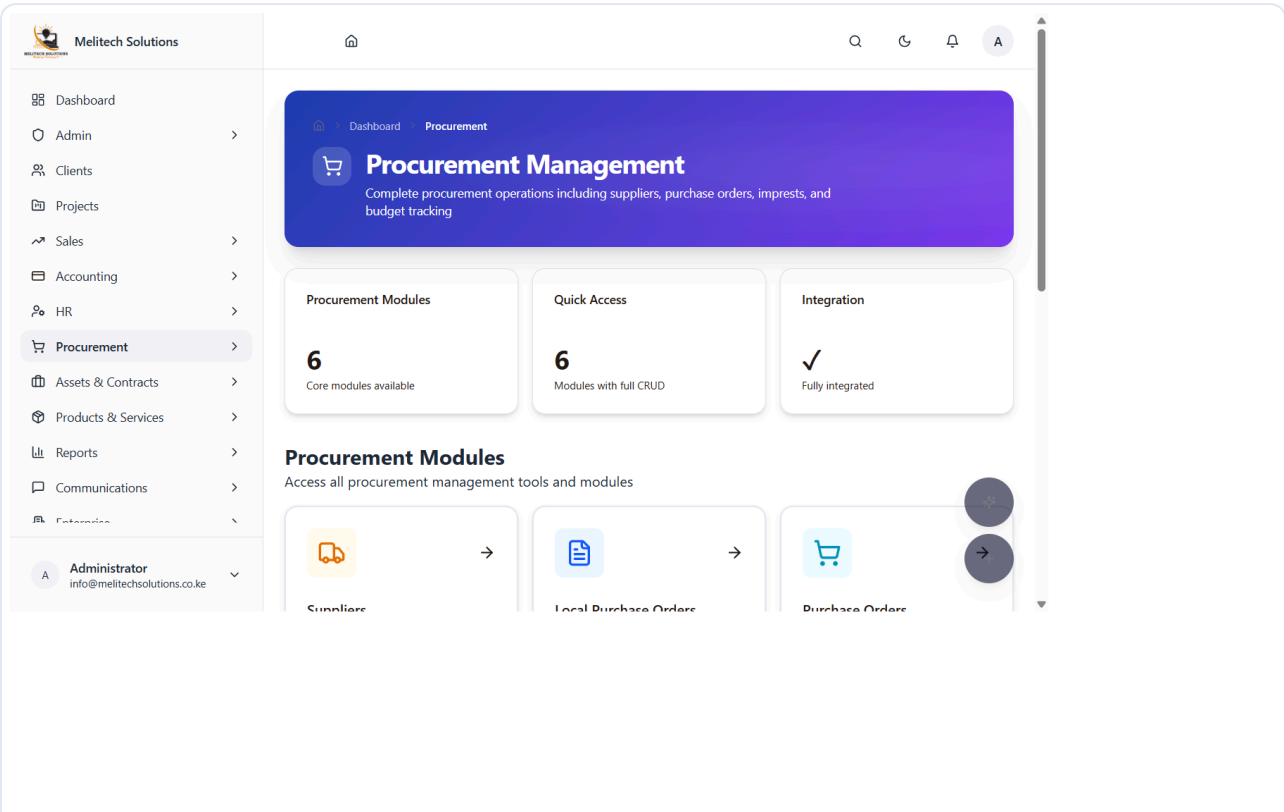


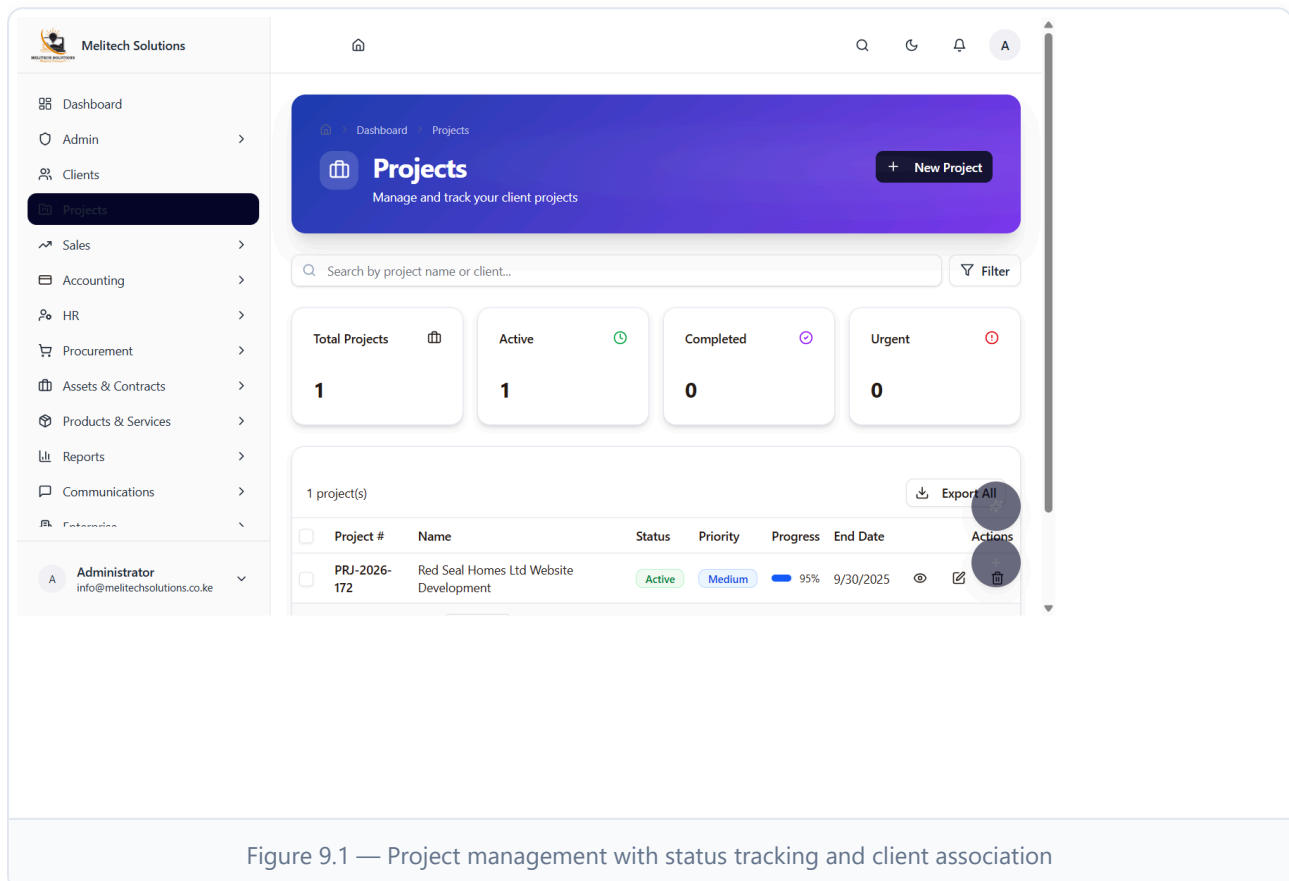
Figure 8.2 — Procurement Hub with unified view of purchasing activities and supplier status

Procurement Sub-Modules

Module	Route	Description
Suppliers	/suppliers	Supplier registry with performance tracking
LPOs (Local Purchase Orders)	/lpos	Purchase order creation and approval
Orders	/orders	Order management and tracking
GRNs (Goods Received Notes)	/grn	Receiving verification against POs
Inventory	/inventory	Stock levels, reorder alerts, warehouse management
Delivery Notes	/delivery-notes	Outgoing delivery documentation
Imprests	/imprests	Petty cash and imprest management

9. Project Management

Track projects from inception to completion with milestones, task assignments, time logging, work orders, and project-specific invoicing.



Features

- **Project Milestones** — Define key deliverables with due dates and completion tracking
- **Time Tracking** — Log hours per team member per task for billable and non-billable work
- **Work Orders** — Issue and track work orders tied to projects
- **Service Invoices** — Generate invoices directly from project time entries and milestones
- **Service Templates** — Reusable service definitions for recurring project types
- **Project Analytics** — Budget vs. actual, timeline adherence, resource utilisation
- **Workflow Automation** — Automated task assignment and status transitions

Feature	Route
All Projects	/projects
Project Milestones	/project-milestones
Time Tracking	/time-tracking
Work Orders	/work-orders
Service Invoices	/service-invoices
Service Templates	/service-templates

10. Products & Services

Catalogue your products and services with pricing, categories, stock management, and linkage to invoices and estimates.

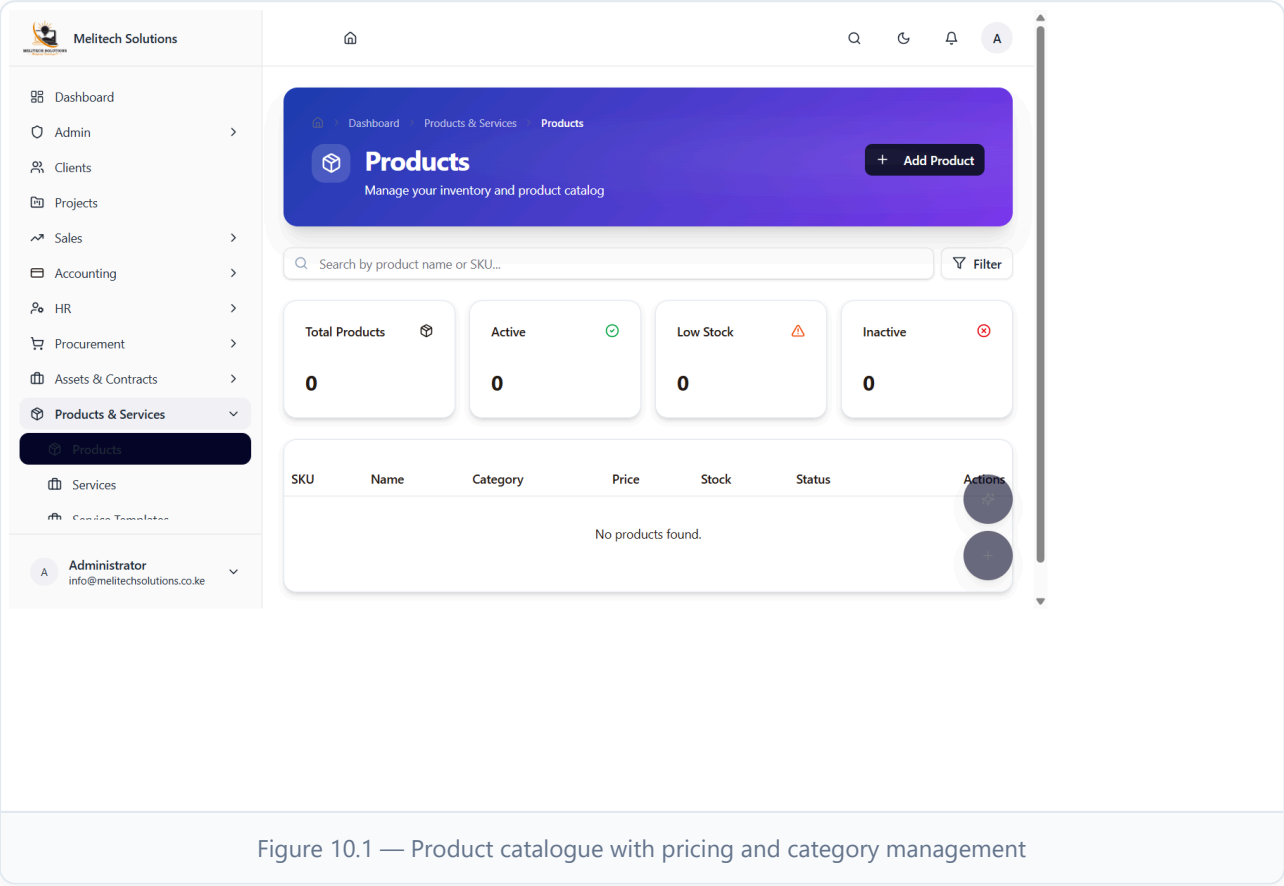


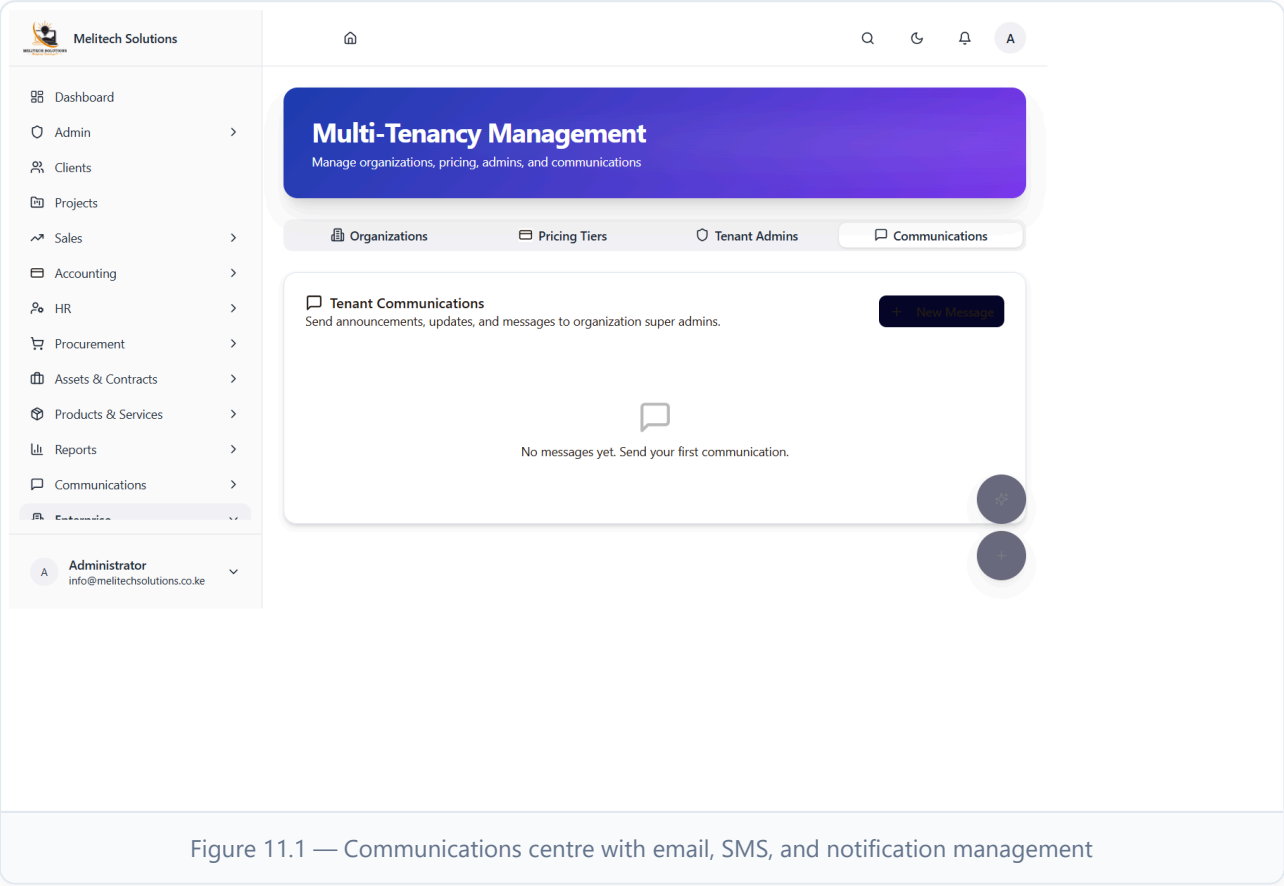
Figure 10.1 — Product catalogue with pricing and category management

Capabilities

Feature	Description
Product Catalogue	Searchable inventory of physical and digital products
Service Catalogue	Service offerings with hourly/fixed pricing
Service Templates	Reusable service bundles for quoting and invoicing
Category Management	Hierarchical categorisation with tags
Pricing Management	Cost price, selling price, markup calculation
Stock Tracking	Quantity on hand, reorder levels, stock alerts
Invoice Integration	Select products/services directly when creating invoices

11. Communications & Messaging

Unified communications hub for email, SMS, in-app notifications, support tickets, team chat, and AI-powered assistance.



11.1 Support Tickets

Internal and client-facing ticket system with priority levels, assignment, SLA tracking, and resolution workflows.

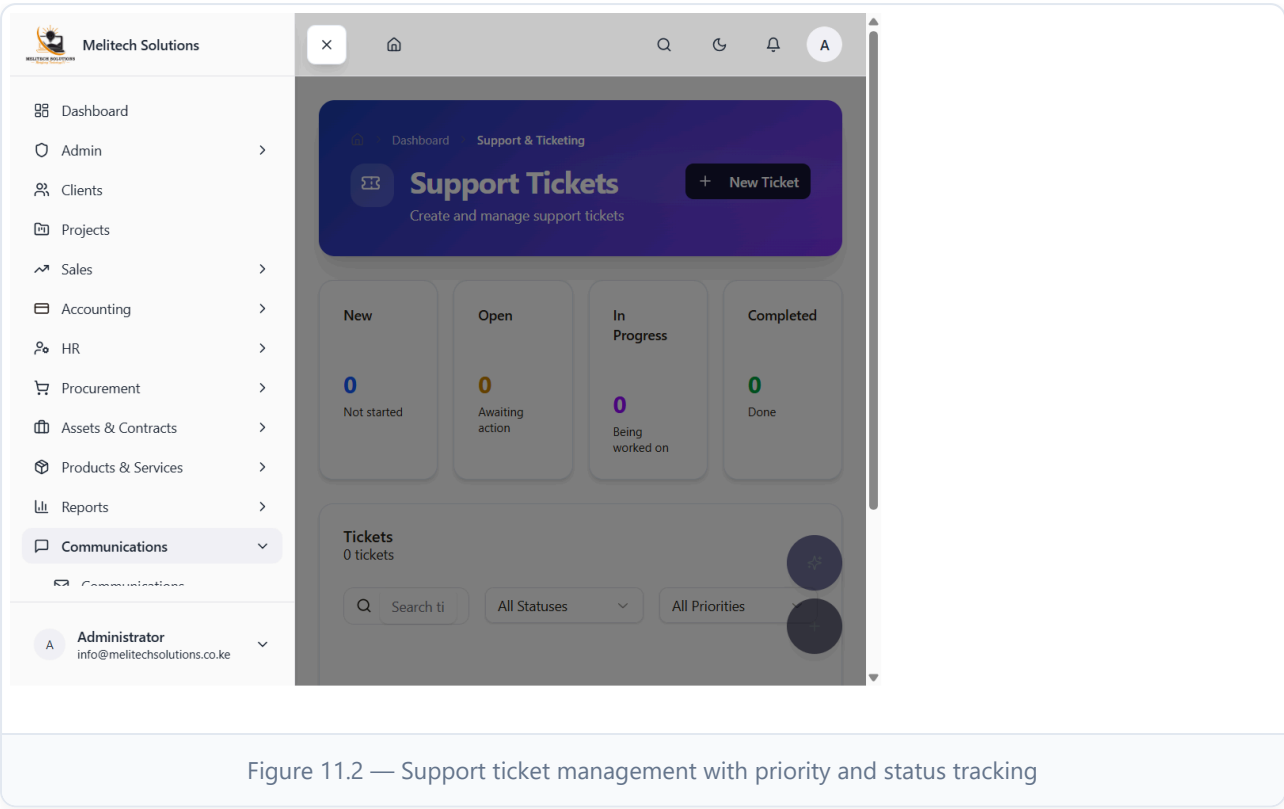


Figure 11.2 — Support ticket management with priority and status tracking

11.3 Document Management

Centralised document repository for storing, organising, and sharing files across all modules. Supports version control and access permissions.

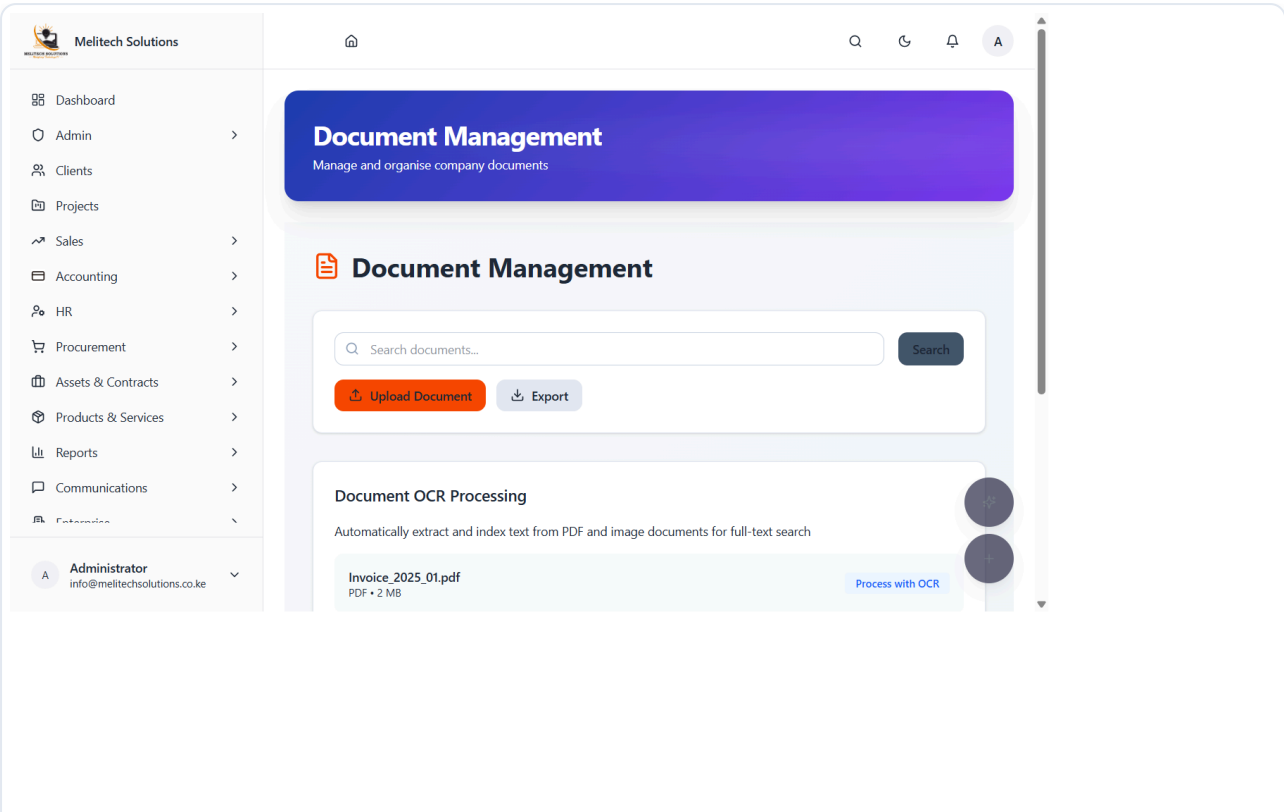


Figure 11.3 — Document management with folder organisation and access controls

Communication Channels

Channel	Route	Description
Communications Hub	/communications	Unified inbox for email and SMS
Support Tickets	/tickets	Issue tracking and resolution
Notifications	/notifications	In-app notification centre
Team Chat	/staff-chat	Real-time messaging between staff
AI Assistant	/ai-hub	AI-powered help and automation
Email Queue	/admin/email-queue	Email delivery queue management
SMS Queue	/admin/sms-queue	SMS delivery queue management

12. Reports & Analytics

Comprehensive reporting across all modules with interactive charts, exportable tables, custom report builder, and KPI dashboards.

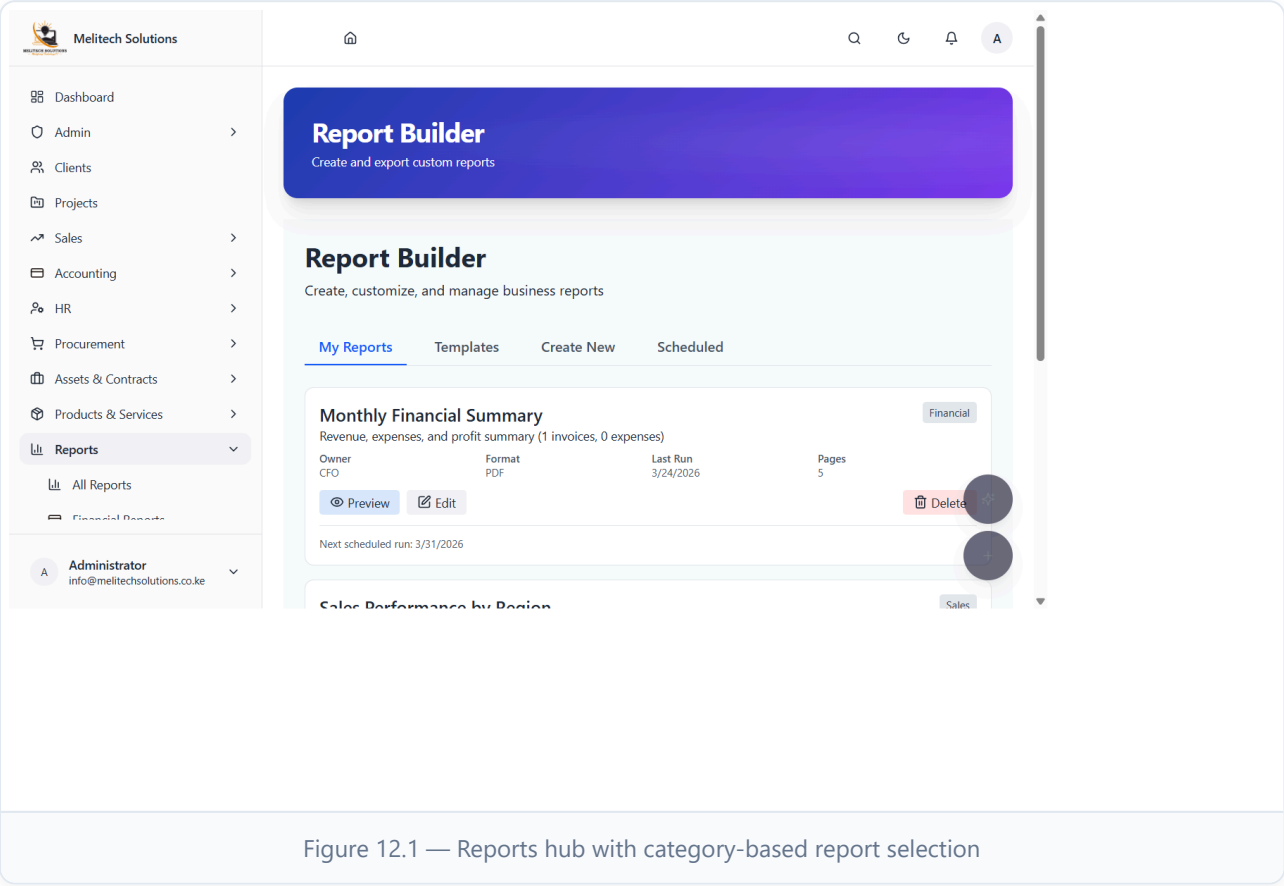


Figure 12.1 — Reports hub with category-based report selection

Report Categories

Category	Reports Available	Route
Financial Reports	P&L, Balance Sheet, Cash Flow, Trial Balance, Tax Reports	/financial-dashboard
Sales Reports	Revenue by period, Client revenue, Pipeline analysis, Conversion rates	/reports/sales
HR Analytics	Headcount, Attrition, Attendance rates, Payroll costs, Leave utilisation	/hr-analytics
Project Analytics	Budget variance, Timeline adherence, Resource utilisation, Profitability	/project-analytics
Advanced Analytics	Custom queries, cross-module data, trend analysis	/advanced-analytics
KPI Tracking	Configurable KPI dashboards with target tracking	/kpi-tracking
Report Builder	Drag-and-drop custom report creation	/report-builder
Revenue Forecasting	AI-assisted revenue predictions	/reports/revenue
Conversion Analytics	Lead-to-client conversion funnel analysis	/reports/conversion
Activity Trail	Full audit log of all system actions	/activity-trail

12.2 Executive Dashboard

High-level business intelligence dashboard with cross-module KPIs, trend charts, and performance summaries for executive review.

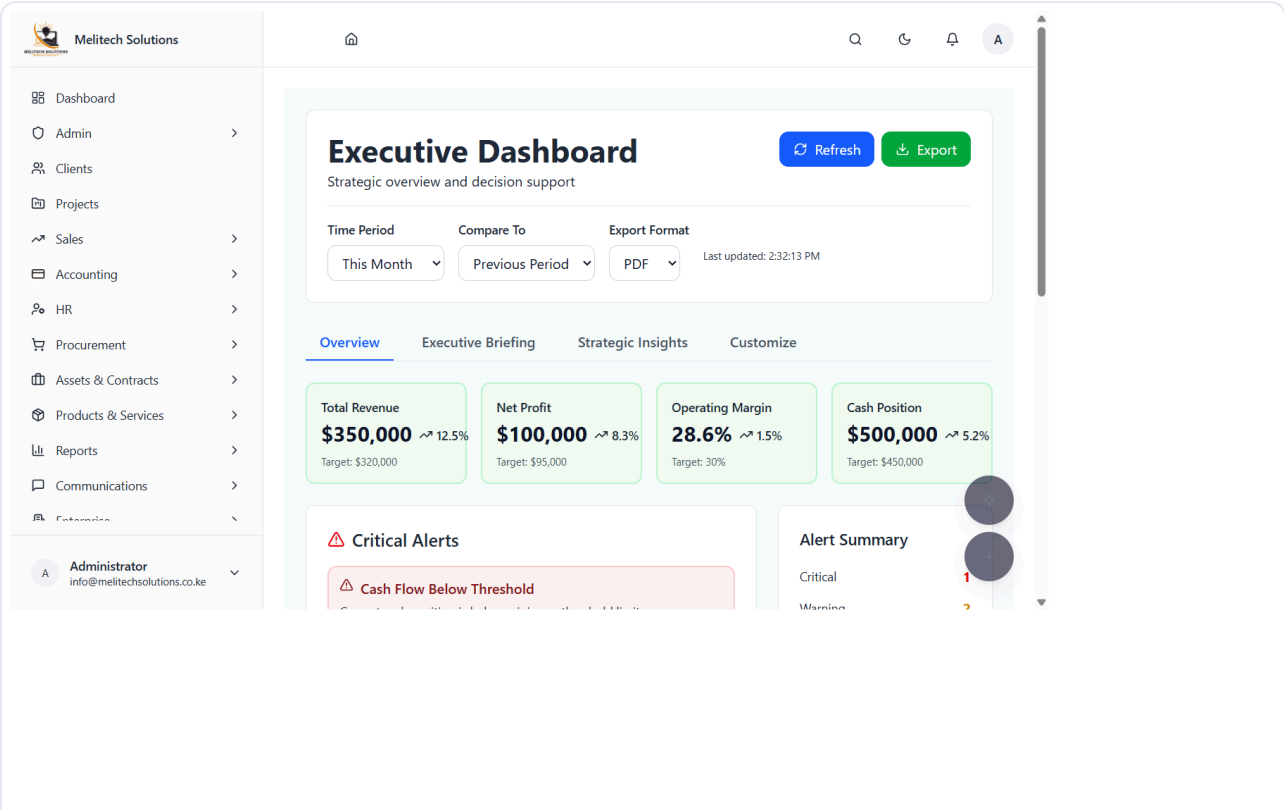
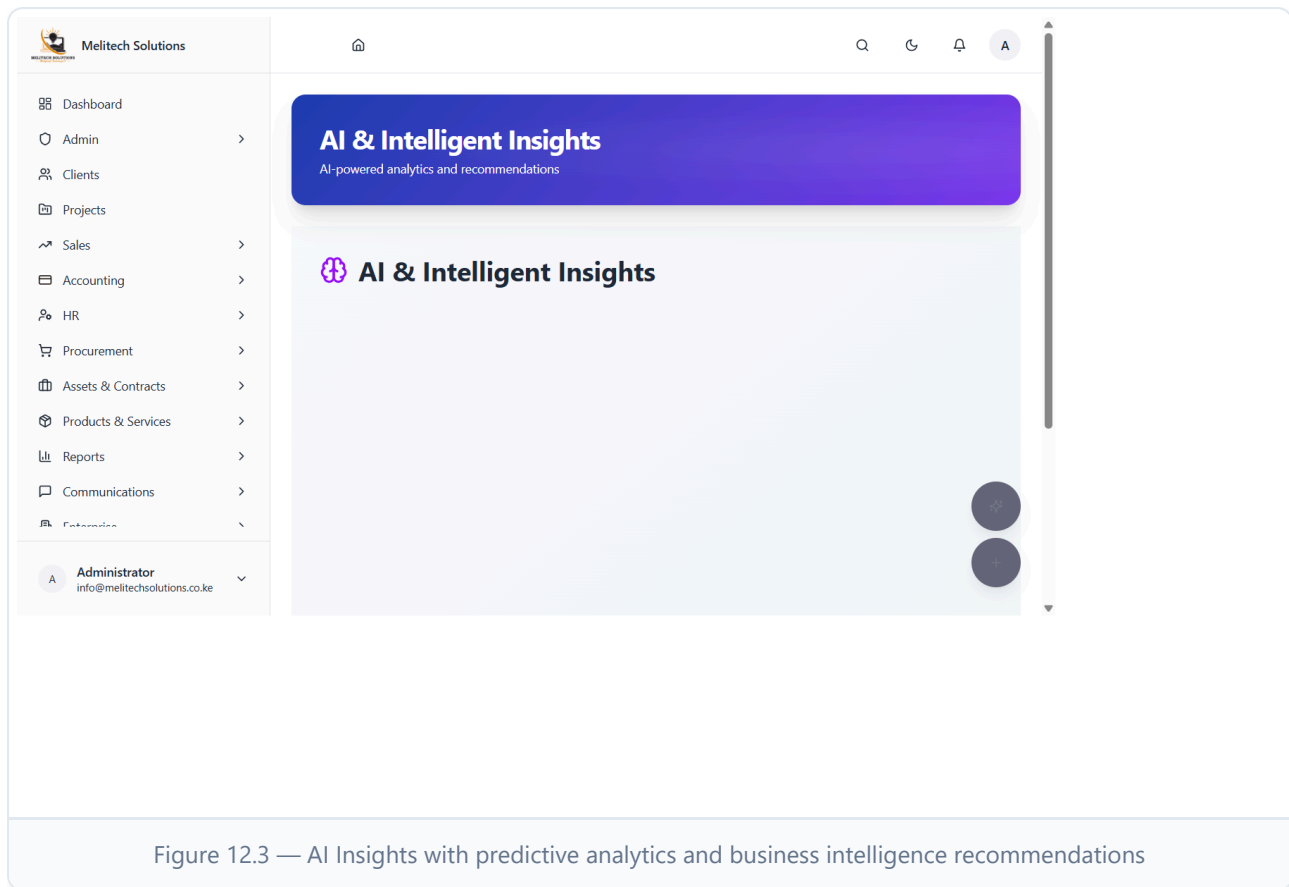


Figure 12.2 — Executive Dashboard with aggregate KPIs and business performance metrics

12.3 AI Insights

AI-powered analytics providing predictive insights, anomaly detection, revenue forecasting, and intelligent recommendations across all business data.



Export Capabilities

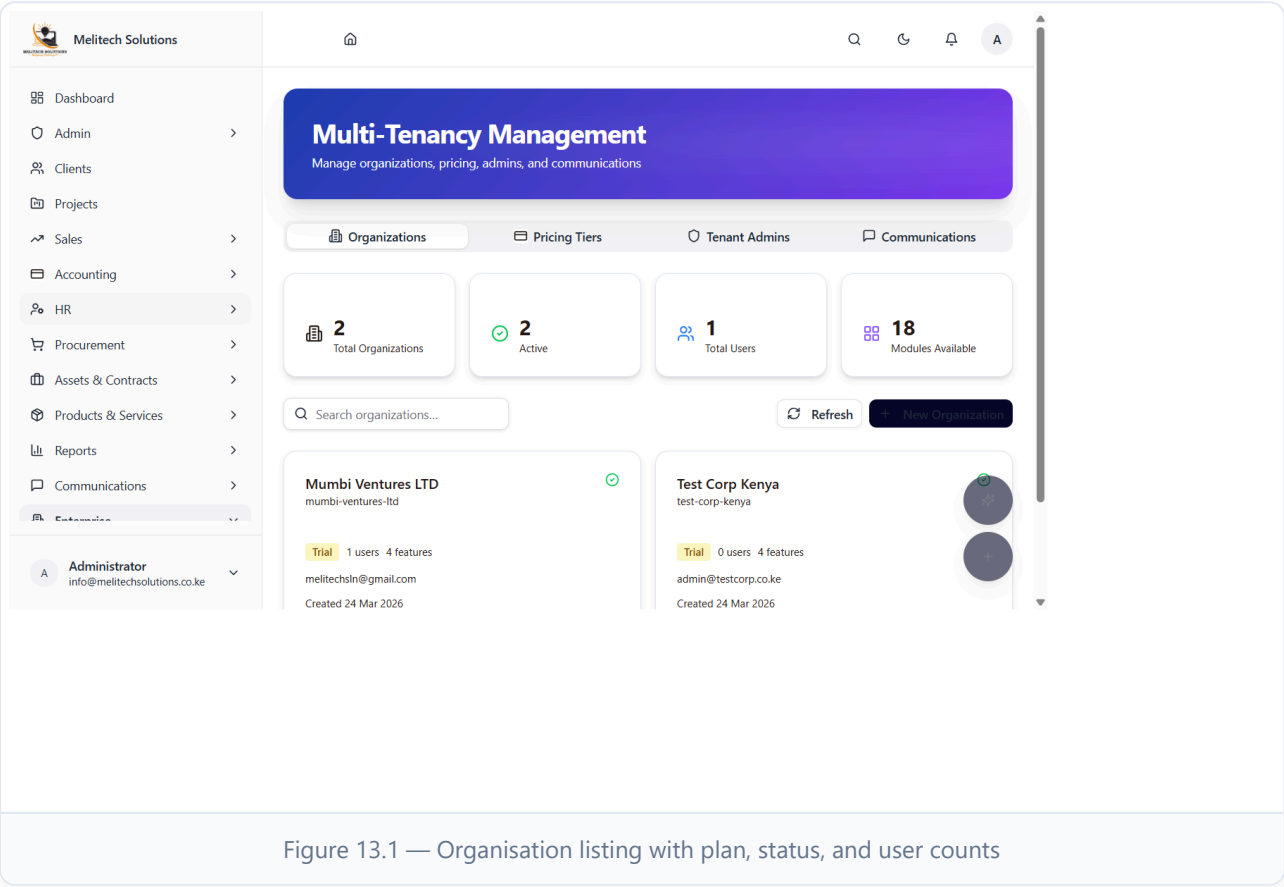
- **CSV Export** — All data tables support one-click CSV download with all columns
- **PDF Export** — Professional PDF reports with branding and charts
- **Print-Ready** — Optimised layouts for direct printing
- **Scheduled Reports** — Configure automatic report generation and email delivery

13. Enterprise Multi-Tenancy

The Enterprise Multi-Tenancy system enables a single Melitech CRM instance to serve multiple organisations — each with isolated data, configurable module access, dedicated super admin accounts, and tailored pricing tiers.

13.1 Organisation Management

Create and manage tenant organisations with dedicated settings, user limits, and module access controls.



13.2 Create Organisation Flow

New organisations are created with a compulsory super admin assignment — either creating a new user or assigning an existing one. This ensures every tenant has an accountable administrator from day one.

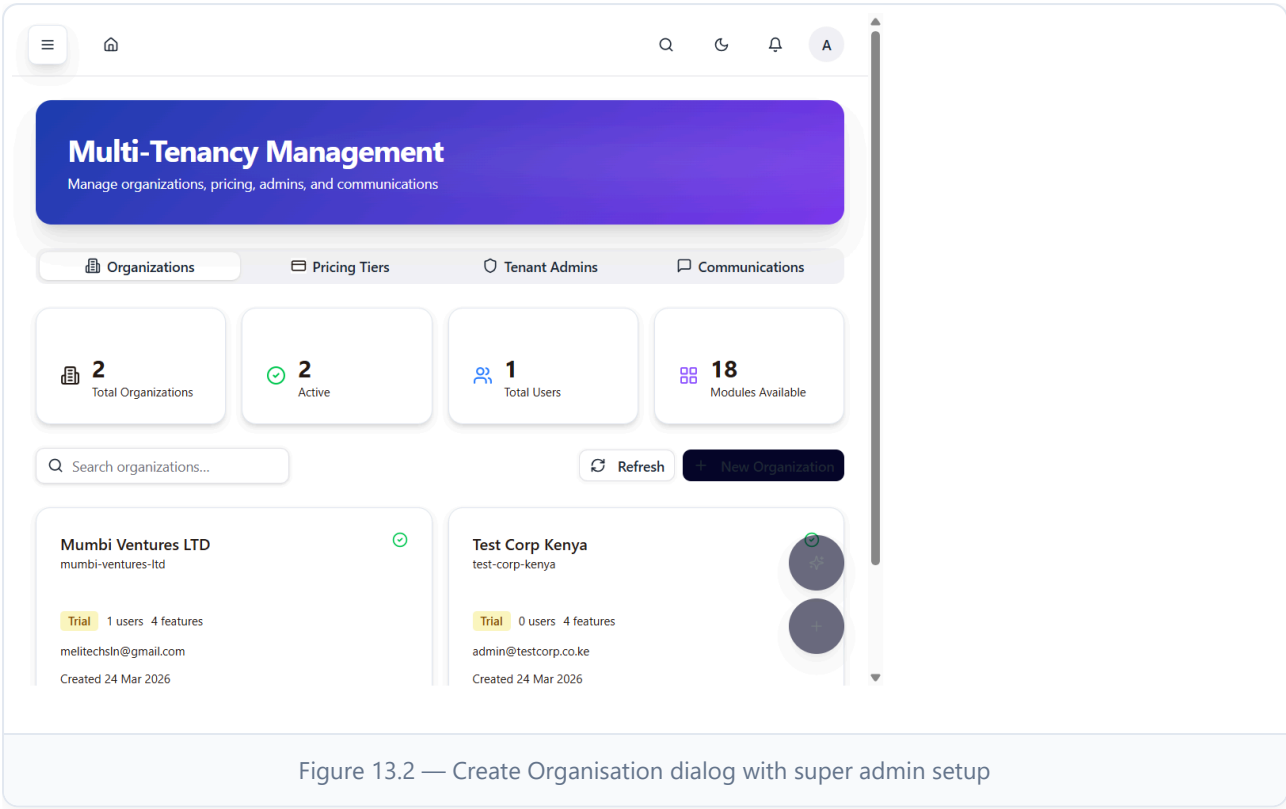
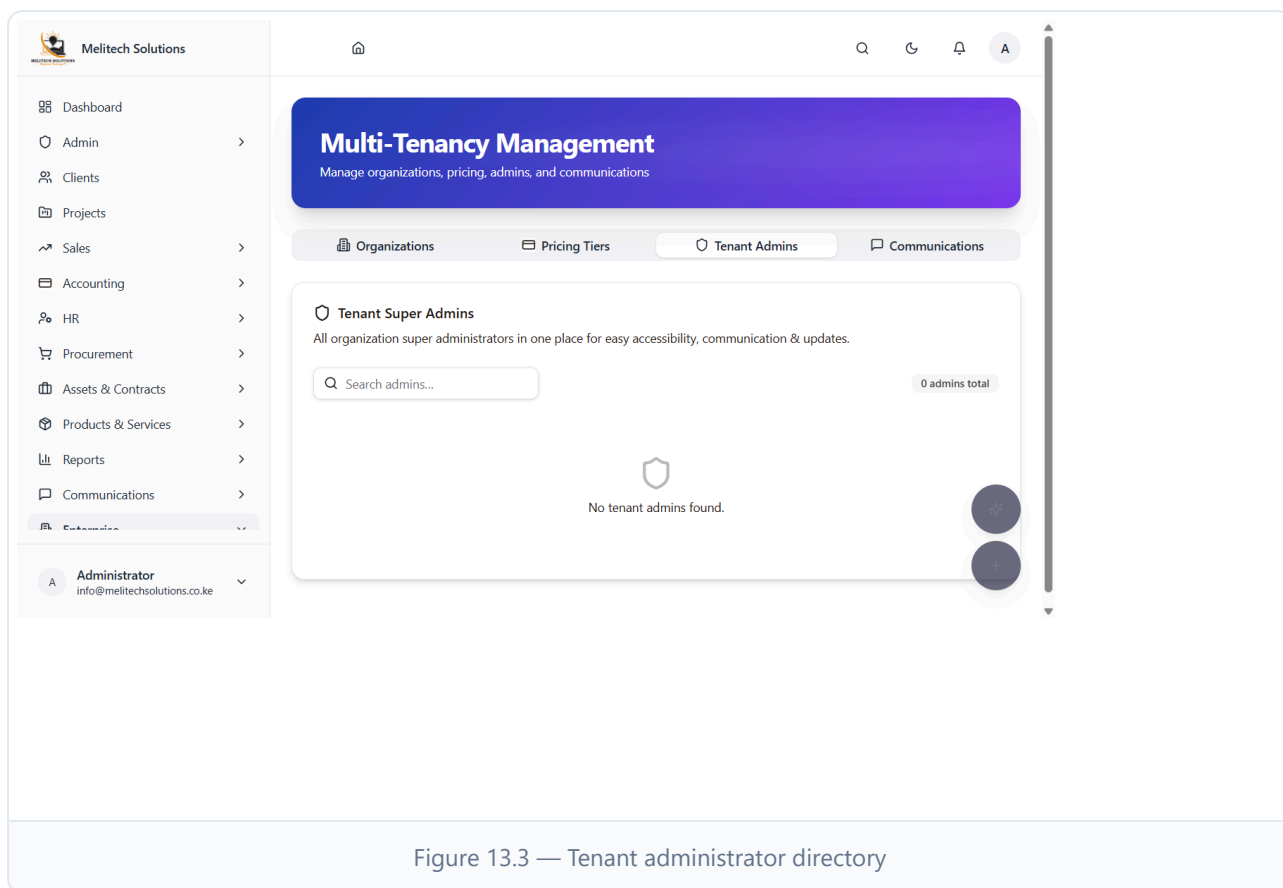


Figure 13.2 — Create Organisation dialog with super admin setup

13.3 Tenant Admin Directory

Centralised view of all tenant administrator accounts with organisation affiliation, contact details, and quick action links.



13.4 Tenant Communications

Broadcast messaging system for platform administrators to communicate with tenant super admins. Supports priority levels (Low, Normal, High, Urgent) and targeting (all admins, specific organisation, or specific user).

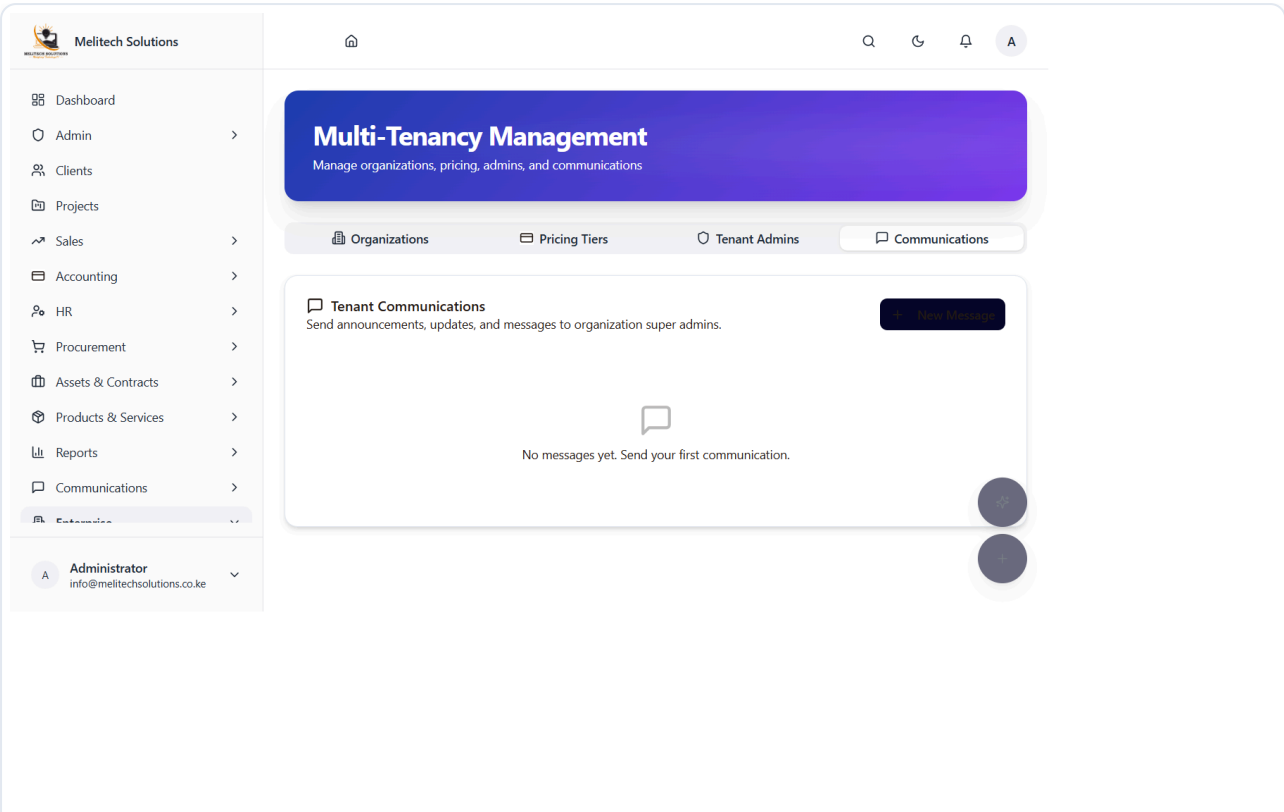
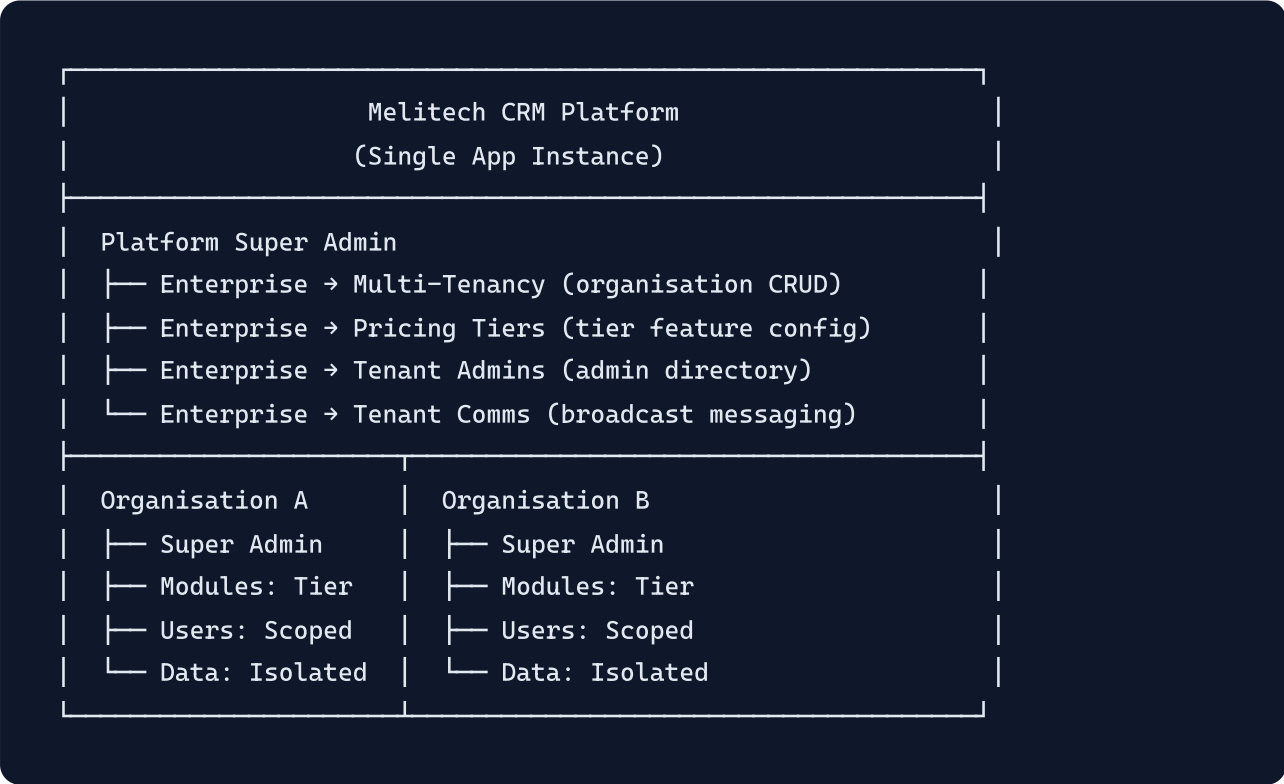


Figure 13.4 — Tenant communications with priority-based messaging

Multi-Tenancy Architecture

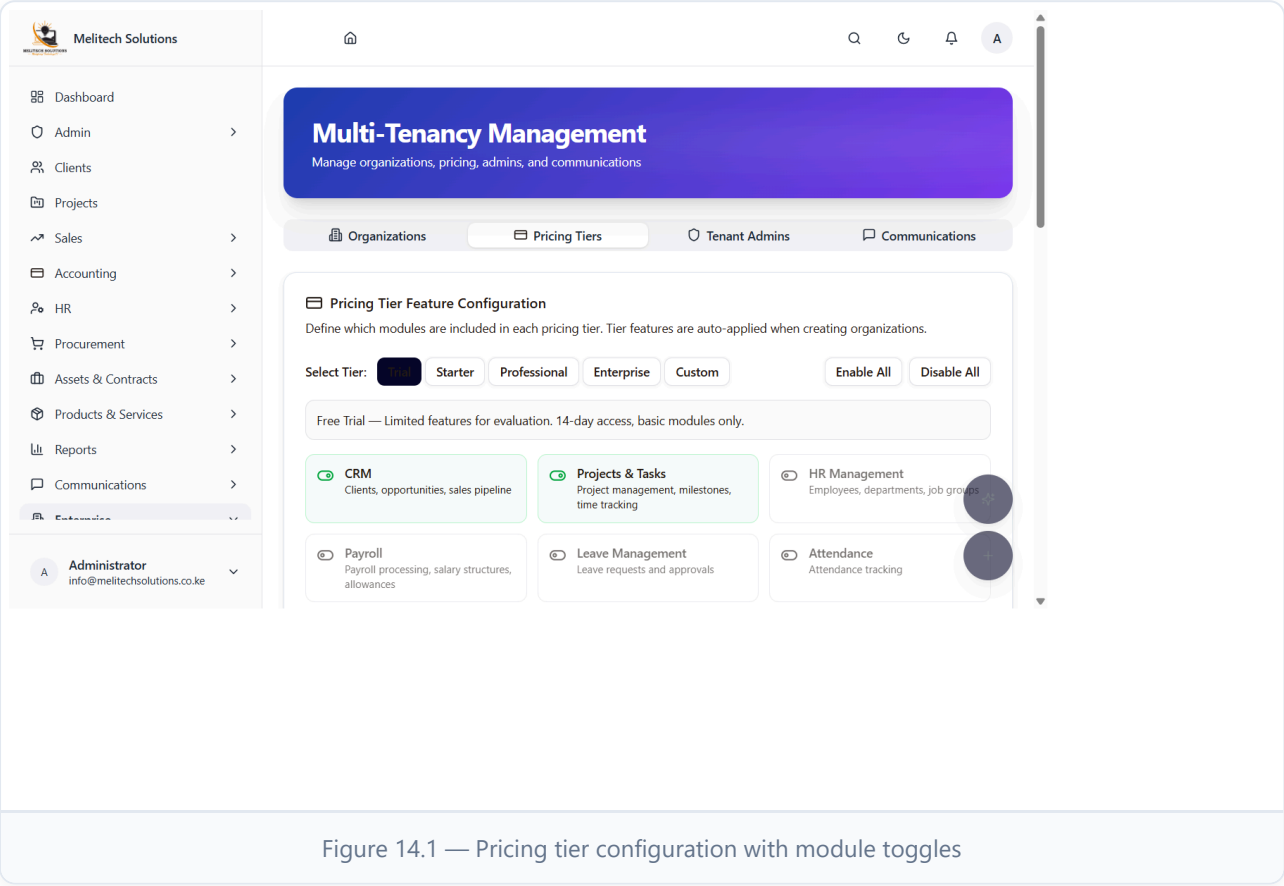


Database Tables

Table	Purpose	Key Columns
organizations	Tenant registry	id, name, slug, plan, isActive, maxUsers, contactEmail
organization_features	Per-org module toggles	organizationId, featureKey, isEnabled, config
pricing_tier_features	Tier-level defaults	tier, featureKey, isEnabled
tenant_messages	Admin communications	senderId, subject, content, priority, targetType

14. Pricing Tiers & Feature Access

Configure which CRM modules are available at each pricing tier. Pricing tiers act as templates — when an organisation is assigned a tier, it inherits the tier's default feature flags.



Tier Comparison

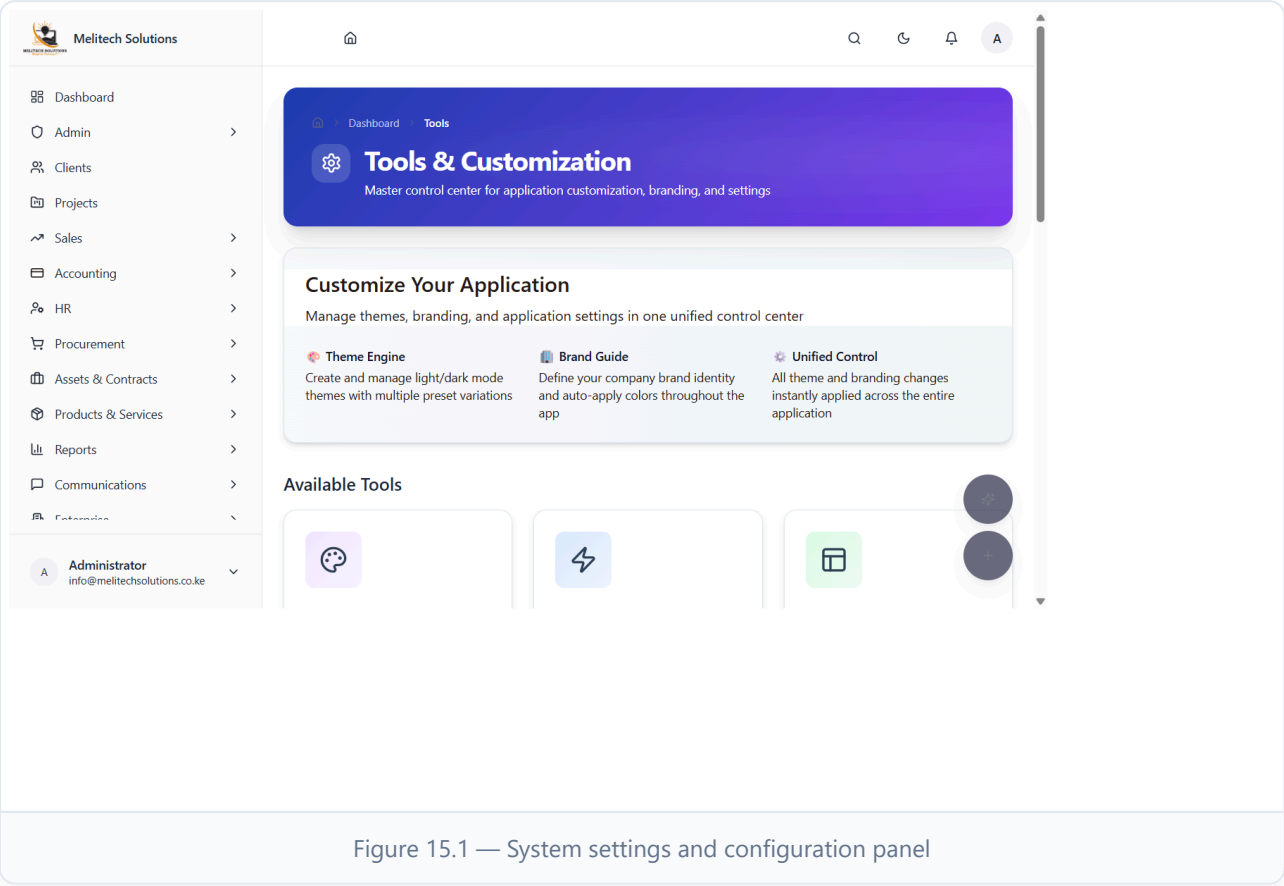
Module / Feature	Trial	Starter	Professional	Enterprise	Custom
CRM (Clients)	✓	✓	✓	✓	✓
Invoicing	✓	✓	✓	✓	✓
Payments	✗	✓	✓	✓	✓
Expenses	✗	✓	✓	✓	✓
Chart of Accounts	✗	✗	✓	✓	✓
HR Management	✗	✗	✓	✓	✓
Payroll (KE Compliant)	✗	✗	✓	✓	✓
Procurement	✗	✗	✓	✓	✓
Project Management	✗	✓	✓	✓	✓
Reports & Analytics	✗	✗	✓	✓	✓
Report Builder	✗	✗	✗	✓	✓
Multi-Tenancy	✗	✗	✗	✓	✓
AI Assistant	✗	✗	✗	✓	✓
Workflow Automation	✗	✗	✗	✓	✓
Custom Fields	✗	✓	✓	✓	✓
Integrations (Stripe/M-Pesa)	✗	✗	✓	✓	✓
Max Users	5	15	50	Unlimited	Custom



Figure 14.2 — Pricing tier feature comparison in the Enterprise panel

15. Tools & Administration

System administration tools for platform configuration, customisation, data management, and monitoring.



15.2 Activity Trail

Complete audit log of all user actions and system events. Track who changed what, when, and from which IP address for compliance and security auditing.

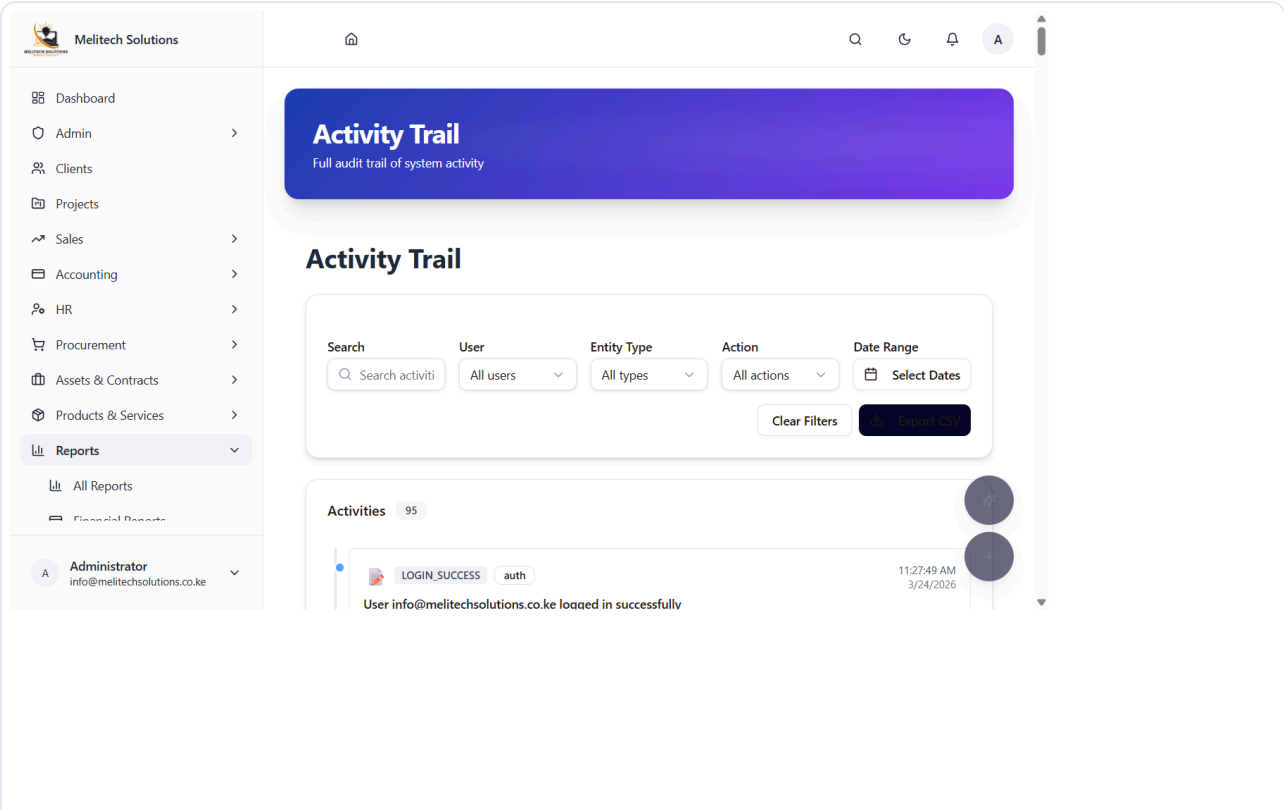


Figure 15.2 — Activity Trail showing user actions, timestamps, and change details

15.3 System Health

Real-time system monitoring dashboard showing server status, database performance, API response times, and resource utilisation.

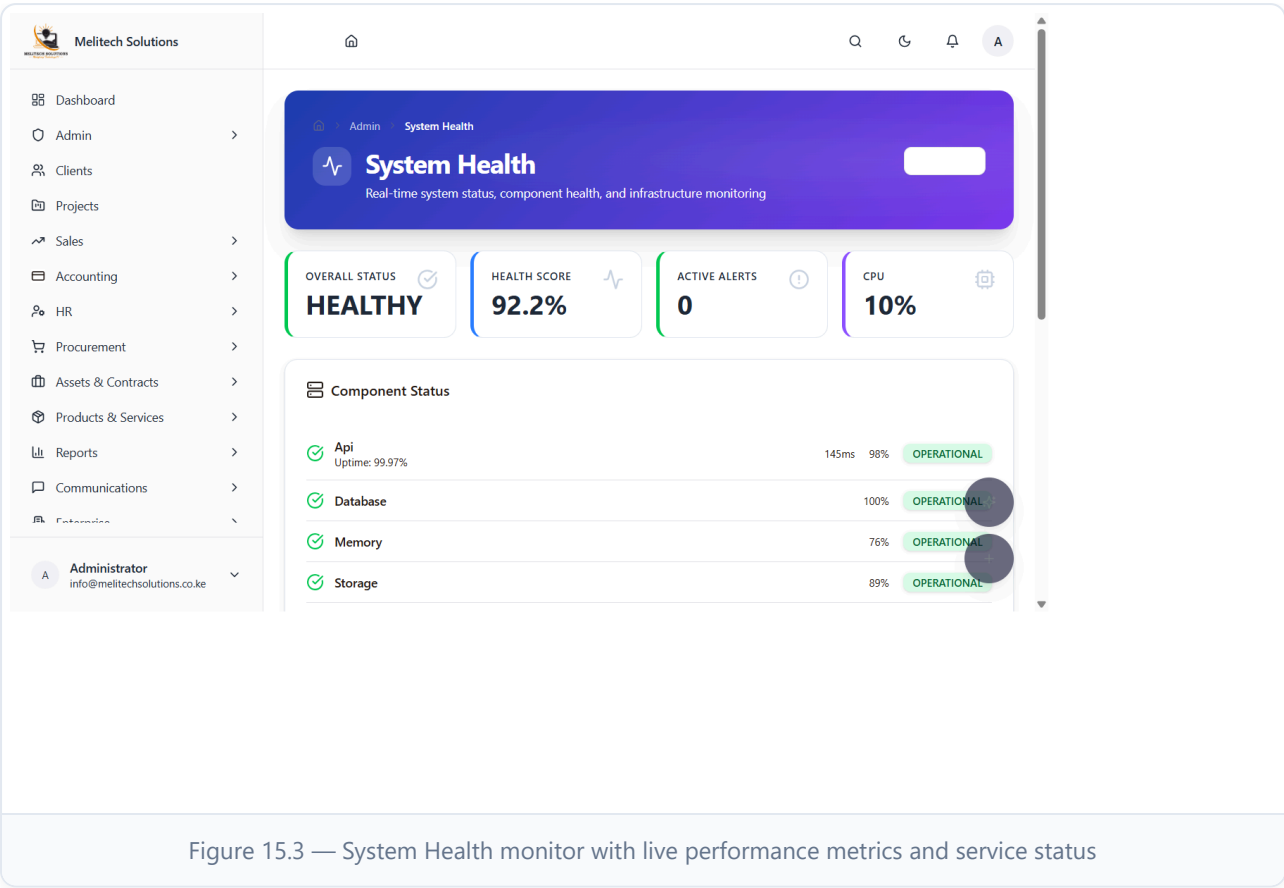


Figure 15.3 — System Health monitor with live performance metrics and service status

Administration Tools

Tool	Route	Description
System Settings	/tools/system-settings	Global configuration: company info, locale, currency, date format
Maintenance Mode	/tools/system-settings	Enable/disable maintenance mode with custom messaging
Backup & Restore	/admin/backups	Database backup, scheduled backups, point-in-time restore
Theme Customisation	/tools/theme-customization	Colour scheme, fonts, sidebar style, dark mode
Brand Customisation	/tools/brand-customization	Logo, company name, branding across all pages
Custom Fields	/tools/custom-fields	14+ field types, entity-specific, validation rules
Homepage Builder	/tools/homepage-builder	Drag-and-drop landing page configuration
Import Data	/import-excel	Excel/CSV import with column mapping
Workflow Automation	/automation	Trigger-based automated actions and notifications
System Health	/system-health	Server status, database health, API response times
Integrations	/integrations	Third-party service connections (Stripe, M-Pesa, email, SMS)
Email Queue	/admin/email-queue	Monitor and manage outgoing email delivery
SMS Queue	/admin/sms-queue	Monitor and manage outgoing SMS delivery
Search	/search	Global search across all modules

16. Security & Authentication

Authentication Flow

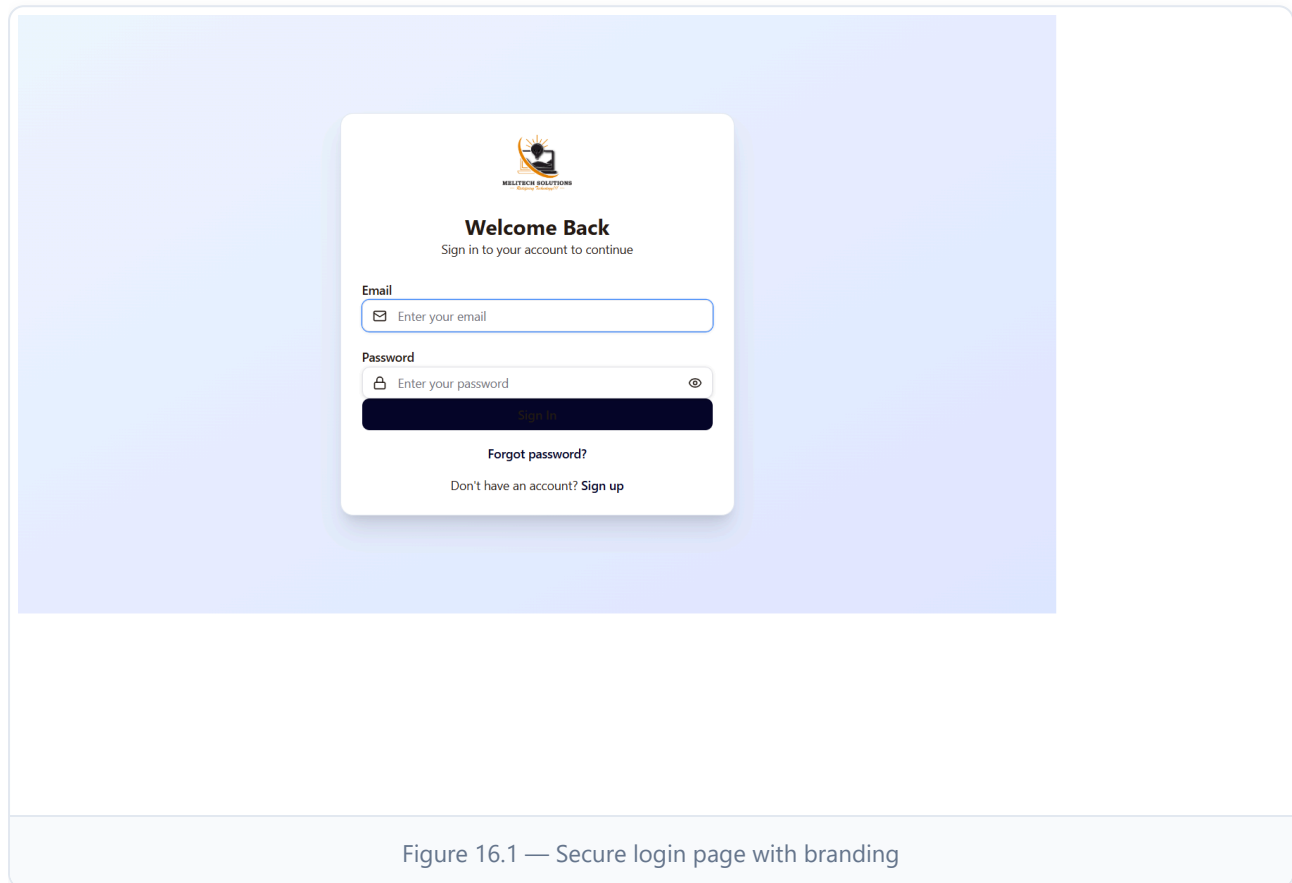


Figure 16.1 — Secure login page with branding

Feature	Implementation
Password Hashing	bcryptjs with 10 salt rounds
Session Management	JWT (JSON Web Tokens) with configurable expiry
Route Protection	AuthGate component wrapping all private routes
CORS Policy	Strict origin validation for cross-origin requests
Input Validation	Zod v4 schemas on all API inputs (server-side)
SQL Injection Prevention	Parameterised queries via Drizzle ORM
XSS Prevention	React DOM escaping + Content Security Policy
Password Reset	Token-based reset with email verification
Session Timeout	Automatic logout after inactivity period
Audit Logging	Activity trail for all state-changing operations

16.2 Security & Compliance Dashboard

Dedicated security management interface for running audits, reviewing access logs, managing encryption settings, and monitoring compliance status.

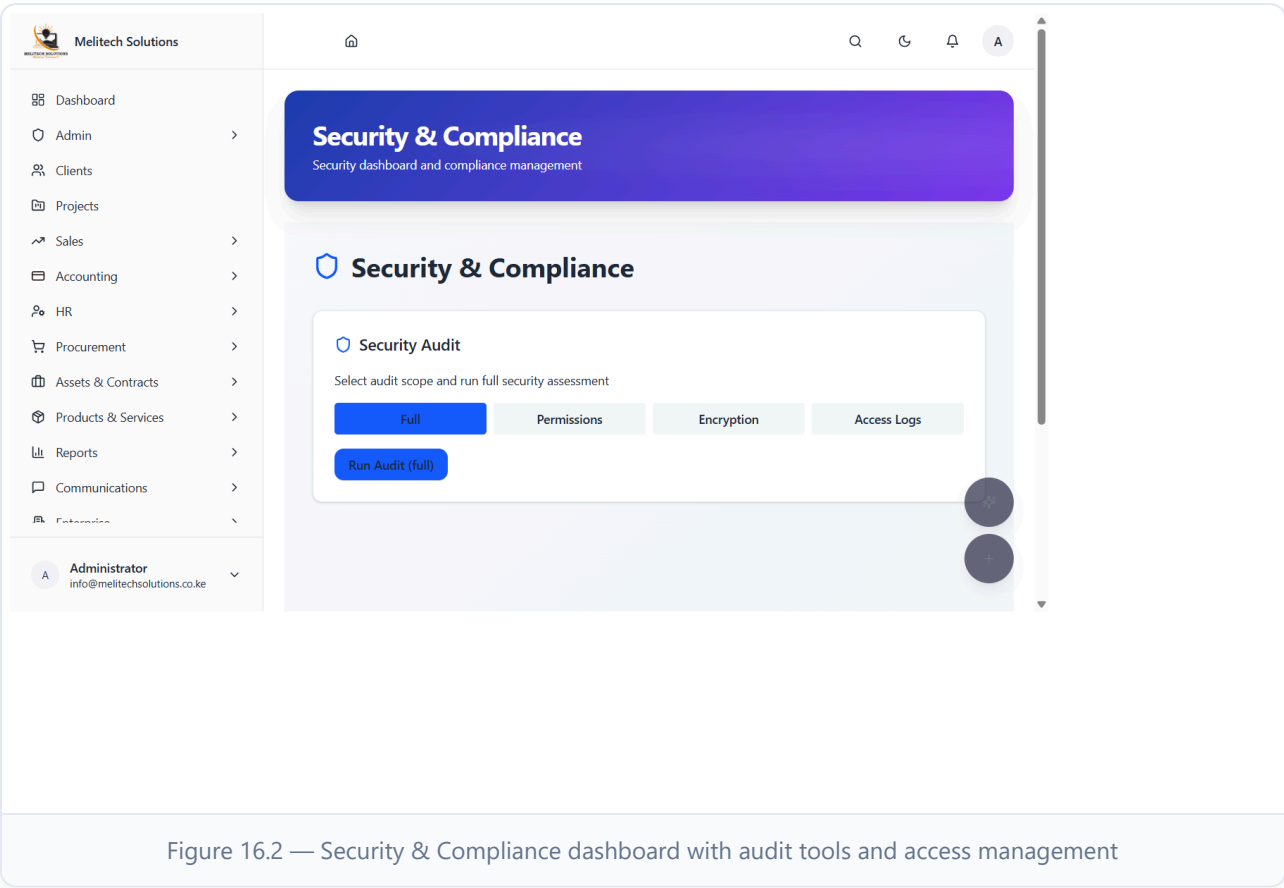


Figure 16.2 — Security & Compliance dashboard with audit tools and access management

Authentication Pages

Page	Route	Purpose
Login	/login	Email + password authentication
Sign Up	/signup	New user registration
Forgot Password	/forgot-password	Password reset request
Reset Password	/reset-password	Token-based password reset

17. Roles & Permissions

Melitech CRM implements granular role-based access control (RBAC) with 150+ frontend permissions and 85+ backend permissions enforced at both the UI and API levels.

System Roles

Role	Access Level	Typical Users
super_admin	Full platform access + Enterprise features	Platform owner, CTO
admin	Full module access, user management	Office manager, department heads
manager	Module-specific management + reports	Sales manager, HR manager
hr	HR module, employees, payroll, attendance	HR officers
accountant	Finance, invoicing, payments, expenses	Finance team
project_manager	Projects, milestones, work orders, time tracking	Project leads
procurement_manager	Procurement, suppliers, LPOs, inventory	Procurement officers
sales_manager	Sales, pipeline, clients, invoices	Sales team leads
staff	Limited access to assigned modules	General staff
client	Client portal: own invoices, projects, tickets	External clients
ict_manager	System tools, tickets, integrations	IT administrators

Permission Enforcement

- **Frontend:** Sidebar items, page access, button visibility all controlled by role
- **Backend:** tRPC middleware checks role array before executing any procedure

- **Feature Flags:** Multi-tenancy feature toggles provide an additional access layer per organisation

18. Integrations & APIs

Melitech CRM integrates with external services for payments, communications, and data exchange.

Payment Integrations

Provider	Type	Capabilities
Stripe	Card Payments	One-time charges, recurring billing, webhook notifications, refunds
M-Pesa (Safaricom)	Mobile Money	STK Push payments, payment confirmations, Kenyan phone validation

Communication Integrations

Service	Purpose	Features
Email (SMTP / SendGrid)	Email notifications	Template-based emails, queue management, delivery tracking
Africa's Talking	SMS notifications	Bulk SMS, delivery reports, Kenyan phone validation

API Architecture

All API endpoints use **tRPC v11** with end-to-end TypeScript type safety. The API supports:

- **108 routers** covering all modules
- **Zod v4** input validation on every procedure
- **JWT Bearer authentication** on all protected endpoints
- **Pagination** with cursor-based and offset-based support
- **Batch requests** via tRPC batching

19. Deployment & DevOps

Docker Deployment

Melitech CRM ships as a Docker multi-container application with the following services:

Container	Image	Port	Purpose
melitech_crm_app	Custom (Node.js)	3000	Application server (frontend + backend)
melitech_crm_db	MySQL 8.0	3307	Database server with persistent volume

Database Migrations

The platform uses Drizzle ORM's migration system with 52 sequential migrations:

Migration Range	Content
0000 – 0030	Core tables (users, clients, invoices, payments, expenses, employees, etc.)
0031 – 0040	Procurement, projects, services, receipts, custom fields
0041 – 0048	Advanced features (approvals, contracts, warranties, attendance)
0049 – 0051	Docker deployment fixes, MySQL syntax corrections, Zod compatibility
0052	Multi-tenancy (tenant_messages, pricing_tier_features)

Environment Configuration

Variable	Description
DATABASE_URL	MySQL connection string
JWT_SECRET	Secret for JWT token signing
STRIPE_SECRET_KEY	Stripe API secret key
STRIPE_PUBLISHABLE_KEY	Stripe publishable key
MPESA_CONSUMER_KEY	M-Pesa API consumer key
MPESA_CONSUMER_SECRET	M-Pesa API consumer secret
EMAIL_HOST / EMAIL_PORT	SMTP server configuration
AT_API_KEY / AT_USERNAME	Africa's Talking SMS credentials
NODE_ENV	production / development

20. Appendix — Complete Module Summary

A. All Routes & Pages

#	Module	Route	Type
1	Landing Page	/	Public
2	Login	/login	Auth
3	Sign Up	/signup	Auth
4	Forgot Password	/forgot-password	Auth
5	Dashboard Home	/crm	Protected
6	Super Admin Dashboard	/crm/super-admin	Protected
7	Admin Dashboard	/crm/admin	Protected
8	HR Dashboard	/crm/hr	Protected
9	Accountant Dashboard	/crm/accountant	Protected
10	PM Dashboard	/crm/project-manager	Protected
11	Clients	/clients	Protected
12	Create Client	/clients/create	Protected
13	Projects	/projects	Protected
14	Create Project	/projects/create	Protected
15	Invoices	/invoices	Protected
16	Create Invoice	/invoices/create	Protected
17	Estimates	/estimates	Protected
18	Quotations	/quotations	Protected

#	Module	Route	Type
19	Receipts	/receipts	Protected
20	Opportunities	/opportunities	Protected
21	Sales Pipeline	/sales-pipeline	Protected
22	Credit Notes	/credit-notes	Protected
23	Debit Notes	/debit-notes	Protected
24	Payments	/payments	Protected
25	Payment Plans	/payment-plans	Protected
26	Expenses	/expenses	Protected
27	Chart of Accounts	/chart-of-accounts	Protected
28	Bank Reconciliation	/bank-reconciliation	Protected
29	Budgets	/budgets	Protected
30	Financial Dashboard	/financial-dashboard	Protected
31	Tax Compliance	/payroll/tax-compliance	Protected
32	Forecasting	/forecasting	Protected
33	Billing	/billing	Protected
34	Employees	/employees	Protected
35	Departments	/departments	Protected
36	Attendance	/attendance	Protected
37	Payroll	/payroll	Protected
38	Leave Management	/leave-management	Protected
39	Job Groups	/job-groups	Protected
40	Performance Reviews	/performance-reviews	Protected

#	Module	Route	Type
41	Suppliers	/suppliers	Protected
42	LPOs	/lpos	Protected
43	Orders	/orders	Protected
44	GRNs	/grn	Protected
45	Inventory	/inventory	Protected
46	Delivery Notes	/delivery-notes	Protected
47	Imprests	/imprests	Protected
48	Products	/products	Protected
49	Services	/services	Protected
50	Service Templates	/service-templates	Protected
51	Assets	/assets	Protected
52	Contracts	/contracts	Protected
53	Warranty	/warranty	Protected
54	Work Orders	/work-orders	Protected
55	Reports	/reports	Protected
56	Report Builder	/report-builder	Protected
57	KPI Tracking	/kpi-tracking	Protected
58	Activity Trail	/activity-trail	Protected
59	Communications	/communications	Protected
60	Tickets	/tickets	Protected
61	Team Chat	/staff-chat	Protected
62	AI Assistant	/ai-hub	Protected

#	Module	Route	Type
63	Multi-Tenancy	/enterprise/tenants	Enterprise
64	Tenant Admins	/enterprise/tenant-admins	Enterprise
65	Pricing Tiers	/enterprise/pricing	Enterprise
66	Tenant Comms	/enterprise/tenant-comms	Enterprise
67	Settings	/settings	Protected
68	Custom Fields	/tools/custom-fields	Protected
69	Theme Customisation	/tools/theme-customization	Protected
70	Import Data	/import-excel	Protected

B. API Router List (Key Routers)

Router	Procedures	Module
auth	login, register, forgotPassword, resetPassword, me	Authentication
clients	list, get, create, update, delete, bulkDelete, export	CRM
invoices	list, get, create, update, delete, send, markPaid	Sales
payments	list, get, create, update, delete, reconcile	Accounting
expenses	list, get, create, update, delete, approve	Accounting
employees	list, get, create, update, delete, getByDepartment	HR
payroll	list, process, getPayslip, kenyanCalculation	HR
attendance	list, clockIn, clockOut, getByDate, report	HR
leave	list, request, approve, reject, getBalance	HR
projects	list, get, create, update, delete, getAnalytics	Projects
suppliers	list, get, create, update, delete	Procurement
products	list, get, create, update, delete	Products
multiTenancy	listOrgs, createOrgWithAdmin, updateOrg, pricingTiers, tenantComms	Enterprise
reports	financial, sales, hr, project, custom	Reports
stripe	createPaymentIntent, webhook, getBalance	Integrations
mpesa	stkPush, callback, getStatus	Integrations

C. Screenshot Index

Figure	File	Description
1.1	16_landing.png	Landing Page
3.1	02_super_admin_dashboard.png	Super Admin Dashboard
4.1	03_clients.png	Clients Management
5.1	04_invoices.png	Invoice Management
6.1	21_accounting.png	Chart of Accounts
6.2	07_payments.png	Payments
6.3	14_expenses.png	Expenses
7.1	06_employees.png	Employees
7.2	15_attendance.png	Attendance
7.3	22_payroll.png	Payroll
7.4	20_leave.png	Leave Management
8.1	11_suppliers.png	Suppliers
9.1	05_projects.png	Projects
10.1	17_products.png	Products
11.1	10_communications.png	Communications
11.2	23_tickets.png	Support Tickets
12.1	12_reports.png	Reports Hub
13.1	08_multitenancy.png	Multi-Tenancy
13.2	24_create_org.png	Create Organisation
13.3	18_tenant_admins.png	Tenant Admins
13.4	19_tenant_comms.png	Tenant Communications

Figure	File	Description
14.1	09_pricing_tiers.png	Pricing Tiers
14.2	25_tier_comparison.png	Tier Comparison
15.1	13_settings.png	Settings
16.1	01_login.png	Login Page

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